

# A VANISHING-CYCLE OBSTRUCTION TO SMOOTHING CALABI–YAU THREEFOLDS

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**ABSTRACT.** We prove a homological obstruction to smoothing projective Calabi–Yau threefolds with nodes and anticanonical cones over smooth del Pezzo surfaces of degrees six and seven. On a crepant resolution that is small over the nodes and blows up the cone vertices, any smoothing forces a relation among link classes bounding in the local Milnor fibers. The relation is nonzero at every node and degree-seven point, and at each degree-six point smoothed through the one-dimensional component. The proof uses vanishing periods and applies to smoothing arcs of arbitrary contact order, extending the necessity direction of Friedman’s criterion. We apply the obstruction to a Batyrev hypersurface whose thirty singularities are individually smoothable but which admits no smoothing, even formally. On its mirror partner, two nodes cannot be smoothed by any analytic deformation, independently of a locally nonsmoothable cone point.

## 1. INTRODUCTION

**1.1. The problem.** Let  $X$  be a projective threefold over  $\mathbb{C}$  with trivial dualizing sheaf and isolated singularities. When every singular germ of  $X$  admits a smoothing, one may ask whether  $X$  itself does. For ordinary double points the answer is classical: by Friedman’s criterion [Fri86], simultaneous smoothing is governed by a relation among the exceptional curves of a small resolution in which every curve appears with nonzero coefficient; see also Rollenske–Thomas [RT09] for a topological formulation in all odd dimensions. Outside the nodal case much less is known. The local deformation theory of the simplest non-nodal germs, the anticanonical cones over del Pezzo surfaces, has been understood since Altmann [Alt95, Alt97], and their smoothing components have explicit geometric descriptions (Gross [Gro97, §5.5]; Petracci [Pet20, Pet19]). Globally, Namikawa proved that a  $\mathbb{Q}$ -factorial Calabi–Yau threefold with isolated toric Gorenstein singularities is smoothable whenever its germs are, and that for a contracted del Pezzo surface of degree 6 the restriction of divisors to the surface determines which smoothing components of the germ are reached by global deformations [Nam97]. What has been missing is a global criterion that couples nodal and divisorial smoothing data on threefolds that need not be  $\mathbb{Q}$ -factorial and carry nodes and del Pezzo cone points at once, and that applies to actual smoothing arcs.

The first three papers of this program [Wue26a, Wue26b, Wue26c] study the isolated Gorenstein toric threefold germs that arise on Batyrev hypersurfaces from unit-edge polygons. They use a local Minkowski-sum obstruction to smoothing and classify the germs with small edge vectors [Wue26a], and, for faces whose germs are smoothable and when a dual-edge-length invariant is at least two, construct deformations that smooth the corresponding points, which give global smoothings under further hypotheses [Wue26b]. They also identify

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a Batyrev mirror pair, associated with a 22-vertex reflexive 4-polytope  $\Delta_{\mathbb{F}_1}$  and its polar, for which both members have only isolated singularities. The remaining smoothing question is *mixed*: nodes and del Pezzo cone points occur on the same threefold, so any smoothing must move both kinds at once. One member carries a cone singularity with no smoothing and is locally obstructed. The other, denoted  $X^\circ$  throughout, has 26 ordinary double points, two anticanonical cones over the del Pezzo surface of degree 7, and two over the del Pezzo surface of degree 6: thirty germs, every one of them locally smoothable. A purely nodal analysis cannot decide  $X^\circ$ : two of its twenty-six nodes are *coloops* of the relation matrix, meaning that their coordinate vanishes on every relation among the classes of the exceptional curves of a small resolution, so no nodal relation is nonzero there and any smoothing would have to recruit the divisorial deformation directions.

**1.2. Results.** Fix the following setting.  $X$  is a connected compact complex threefold with  $\omega_X \cong \mathcal{O}_X$  and  $h^1(\mathcal{O}_X) = 0$  whose singular points are ordinary double points and anticanonical cones over del Pezzo surfaces of degrees 6 and 7;  $\pi: \hat{X} \rightarrow X$  is a crepant resolution, small over the nodes, and near each cone point  $p$  isomorphic to the blowup of the vertex, so that  $\pi^{-1}(p)$  is a single exceptional surface  $E_p$ , a copy of the del Pezzo surface of the cone. For a singular point  $p$  let  $L_p$  denote its link. A *one-parameter deformation* of  $X$  is a flat proper analytic family  $\mathcal{X} \rightarrow \mathbb{D}$  over a disk  $\mathbb{D}$  with  $\mathcal{X}_0 \cong X$ , and it is a *one-parameter smoothing* if  $\mathcal{X}_s$  is smooth for all  $s \neq 0$ . Such a smoothing induces at each singular point a smoothing of the local germ, with Milnor fiber  $\mathcal{M}_p$ , and we set

$$\mathcal{R}_p := \ker(H_2(L_p; \mathbb{Q}) \rightarrow H_2(\mathcal{M}_p; \mathbb{Q})),$$

the part of the link homology that bounds in the local Milnor fiber. By Altmann's computation, recalled in Section 2.2, the miniversal base of a node or of a degree-7 cone germ has one reduced component, and that of a degree-6 cone germ has two, of dimensions 1 and 2; a smoothing sends the germ into one of them, and  $\mathcal{R}_p$  depends on which.

**Theorem A** (mixed necessity; = Theorem 6.1). *Suppose  $X$  as above admits a one-parameter smoothing. Then the space*

$$K(X) := \left\{ \kappa \in \ker \left( \bigoplus_p H_2(L_p; \mathbb{Q}) \rightarrow H_2(\hat{X}; \mathbb{Q}) \right) : \kappa_p \in \mathcal{R}_p \text{ at every cone point } p \right\}$$

*contains, for every node  $q$ , for every degree-7 cone point  $q$ , and for every degree-6 cone point  $q$  whose induced local smoothing lies in the one-parameter component of its miniversal base, a class  $\kappa$  with  $\kappa_q \neq 0$ . Here the maps are induced by the inclusions  $L_p \subset X \setminus \text{Sing } X \subset \hat{X}$ .*

The subspaces  $\mathcal{R}_p$  are computed in closed form (Proposition 2.1): at a degree-7 point  $\mathcal{R}_p$  is the unique line spanned by a  $(-2)$ -class of  $K^\perp \subset H_2(E_p; \mathbb{Q})$ , and at a degree-6 point it is the  $A_1$  or the  $A_2$  summand of the root lattice  $K^\perp \cong A_1 \oplus A_2$ , according to the component of the local miniversal base through which the family smooths the germ. These subspaces coincide with the lattices  $\Xi(V)$  attached by Kuznetsov–Prokhorov [KP23, Theorem 1.9] to the del Pezzo threefolds  $V$  that sweep out the local smoothings.

Theorem A is a finite linear-algebra test on any concrete  $X$ . It is finite although  $\mathcal{R}_p$  refers to a smoothing whose existence is in question, because the degree-6 points contribute only finitely many choices of component: one runs the test on each, as Section 9.1 does. On the distinguished hypersurface every choice obstructs.

**Theorem B** (= Theorem 6.5 and Corollary 6.7). *Let  $\Delta_{\mathbb{F}_1}$  be the 22-vertex reflexive polytope printed in Table 3, with polar  $\Delta_{\mathbb{F}_1}^\vee$  printed in Table 1. Let  $X^\circ$  be an anticanonical hypersurface,*

regular in Batyrev's sense [Bat94, Def. 3.1.1], in the Gorenstein Fano toric fourfold whose fan is the face fan of  $\Delta_{\mathbb{F}_1}^\vee$ , equivalently whose anticanonical polytope is  $\Delta_{\mathbb{F}_1}$ . Then  $X^\circ$  admits no one-parameter smoothing. Moreover no fiber of a small representative of the miniversal deformation of  $X^\circ$  is smooth, and no formal arc in the miniversal deformation of  $X^\circ$  meets all thirty local smoothing loci, in the sense of Corollary 6.7.

Theorems A and B are the main results. The first is the general topological necessity statement; the second is its application to the mixed Batyrev hypersurface that motivated the criterion.

Call a Batyrev mirror pair *doubly isolated* when both members have only isolated singularities. The mirror partner  $X_\Delta$  carries an anticanonical cone over  $\mathbb{F}_1$ , a germ with no smoothing [Wue26a, Wue26c], so both members of the pair are non-smoothable. The criterion says more. Extended to deformations that leave such a germ in place (Section 7), it shows that  $X_\Delta$  carries the global obstruction as well: two of its twenty nodes are smoothed by no one-parameter deformation at all, and this is independent of the  $\mathbb{F}_1$  germ, which already prevents a smoothing for a purely local reason. The two members of the pair are therefore not an instance of one obstruction seen locally and globally; each fails globally, and one of them fails locally in addition.

The analytic core of Theorem A is quantitative: the period of the holomorphic 3-form of the nearby fiber over the vanishing cycle of a node equals the local smoothing parameter times a unit (Lemma 3.1; for the linearity of the nodal period see also [RT09, (1.4)]), and an analogous nonvanishing holds for the distinguished cycles of the degree-7 and one-parameter degree-6 smoothings (Lemma 3.3), where the decisive constant is an explicit period equal to  $4\pi^2$  on a cycle we construct by hand. Consequently a smoothed germ cannot have rationally null-homologous vanishing homology, at any order of tangency of the family with the local discriminants. In the purely nodal case this yields:

**Corollary 1.1** (Corollary 6.3). *For a connected compact complex threefold with  $\omega \cong \mathcal{O}$ ,  $h^1(\mathcal{O}) = 0$  and only ordinary double points, any one-parameter smoothing forces, for every node  $q$ , the existence of a relation among the exceptional curve classes of a small resolution that is nonzero at  $q$ . This is the necessity half of Friedman's criterion. It is not new in the nodal case: [Nam02, Theorem 2.5] proves the equivalence for threefolds with terminal singularities, and [Col25, §4] gives a differential-geometric proof of necessity. What the method adds is that it needs no unobstructedness and no hypothesis of a smooth total space, and that it survives at germs which are not local complete intersections, where those two literatures stop.*

For  $X^\circ$  the exact finite input localizes the obstruction topologically:

**Proposition 1.2** (Proposition 6.4). *In any hypothetical one-parameter smoothing of  $X^\circ$ , the vanishing 3-spheres of the two coloop nodes would be null-homologous in  $H_3(X_s; \mathbb{Q})$ .*

Lemma 3.1 says that a smoothed node has a nonzero vanishing-cycle class, whereas Proposition 1.2 makes that class zero at either coloop node. This contradiction proves Theorem B. A complementary second proof, with a different period input, runs through a degree-7 point instead of the nodes (Section 6).

The one-germ result and the bounded census are applications of the general criterion rather than additional main theorems. On the seven-vertex polytope  $\Delta_7$ , the unique singularity is a degree-7 cone and its root line survives in  $H_2(\hat{X}; \mathbb{Q})$ , so Corollary 9.1 obstructs a smoothing. Among the 77 admissible polytopes with at most nine vertices and a pentagonal 2-face, the same test obstructs 76 (Proposition 9.2).

Section 8 interprets the matrix used in the finite test. Choose a basis of the link homology at each germ. Each basis class defines a linear functional on the ambient Picard group by intersection with its image on the resolution; its values on ambient divisor classes form a *row* of the matrix  $B$ . Read in local ray coordinates, the rows at a germ form a rational basis whose complexification is  $T^1$  (Proposition 8.2). For the isolated hypersurfaces under consideration, vanishing of Batyrev’s correction term implies that the ambient divisors detect the entire topological relation kernel (Proposition 8.3). Finally, under the resolution hypotheses of Proposition 8.4, algebraic  $\mathbb{Q}$ -factoriality of  $X$  forces every divisorial row to vanish. This explains compatibility with Gross’s smoothing theorem for primitive type II contractions [Gro97, Theorem 5.8].

The criterion does not obstruct every mixed example. The anticanonical hypersurface  $X_{19}$  of a 19-vertex reflexive polytope from the companion paper [Wue26c] has fourteen nodes and one degree-7 cone point, and Section 9.5 shows that it again has two nodal coloops. Applied to  $X_{19}$ , the finite test is silent: no coordinate to which Theorem A’s nonvanishing conclusion applies is forced to vanish (Section 9).

**Example 1.3.** The companion paper [Wue26e] proves that  $X_{19}$  is smoothable and that its full deformation space is smooth of dimension 30. Every smoothing must deform its degree-7 cone point as well as its nodes; the nodal relation space alone does not satisfy the necessity condition.

**1.3. Method.** The general necessity theorem has two inputs. First, the vanishing-period computations of Section 3 convert nonvanishing of local smoothing parameters into nonvanishing homology classes. Second, the local-to-global membership statement requires a boundary-marked topological comparison. The germ Milnor fiber is identified with a del Pezzo threefold complement as a manifold with boundary, compatibly with the two natural circle fibrations of the link (Lemma 4.2); together with the classification of del Pezzo threefolds of degrees 6 and 7 (Fujita [Fuj80]), this computes the subspaces  $\mathcal{R}_p$ . Surgery along the links and Mayer–Vietoris then place the boundary of every 3-cycle of  $X_s$  in the space  $K(X)$  of Theorem A. Its application to  $X^\circ$  adds a third input: an exact integral computation on a fixed crepant resolution shows that on the relevant restricted kernels the coordinates of the two coloop nodes and of the degree-7 directions vanish identically (Section 5); all decisive matrix data are printed, and Appendix A lists the programs that recompute them.

The argument does not use the first-order deformation theory of  $X$ : the localization map and the comparison of coordinates on local deformation spaces do not enter. The periods apply directly to the smoothing arc, including arbitrary orders of contact with the local discriminants. For nodes, the existence criterion for an actual smoothing is already stated in [RT09, Thm. 1.3]. The point here is to obtain the mixed necessity statement directly from periods, without assuming unobstructedness of the deformation space of  $X$ .

**1.4. Relation to prior work.** Friedman [Fri86] established the relation criterion for first-order smoothings of nodal Calabi–Yau threefolds. Rollenske–Thomas [RT09, Thms. 1.2–1.3] recall both that statement and the criterion for actual smoothings obtained using unobstructedness, and develop its higher-dimensional counterpart. Namikawa [Nam02, Theorem 2.5] proved the nodal statement as an equivalence, for terminal singularities, and Collins [Col25, §4] proves necessity there by a differential-geometric argument. In the opposite direction, Aguilar [Agu26] shows that for log conifold transitions of Fano threefold pairs of index two the analogous balancing conditions vanish and local first-order smoothings of nodes lift independently, so no relation is forced; the contrast marks how much of the present obstruction

is Calabi–Yau specific. Ning [Nin26] extends local-freeness results beyond the local complete intersection case. Friedman’s generalized  $T^1$  lifting theorem likewise permits isolated Du Bois singularities that are not local complete intersections [Fri26, Thm. 3.10]. The companion paper [Wue26e] combines that theorem with crepant partial models and analytic blow-down to prove integrability of the selected deformation spaces under its stated projection hypothesis. Friedman–Laza [FL25, §5] identify the first-order localization data of Calabi–Yau varieties with isolated local complete intersection singularities in terms of link homology, and [FL24] develops the corresponding local theory; our Section 5 uses only the resolution-side avatar of that picture, in the form of an integral matrix. Altmann [Alt95, Alt97] computed the miniversal deformations of the del Pezzo cone germs and their smoothing components, which Gross [Gro97, §5.5] describes geometrically; Petracci [Pet20, Pet19] and Kaloghiros–Petracci [KP21] give modern treatments, the latter computing the Betti numbers of the degree-6 Milnor fibers, in agreement with Section 2. The classification of the sweeping del Pezzo threefolds is due to Fujita [Fuj80]; we follow Fujita in recording that the degree-6 case of Iskovskikh’s earlier list omits  $\mathbb{P}(T_{\mathbb{P}^2})$ , and cite Fujita, with Kuznetsov–Prokhorov [KP23] as the modern reference. On the constructive side, multi-parameter deformations of toric varieties in a single character degree were built by Ilten–Vollmert [IV12]. The companion paper [Wue26e] proves the mixed smoothing criterion, including the two-parameter degree-six component, and proves the smoothability of  $X_{19}$  in Example 1.3.

**Convention 1.4.**  $\mathrm{dP}_d$  denotes a del Pezzo *surface* of degree  $d = K^2$ ; thus  $\mathrm{dP}_6 = \mathrm{Bl}_3 \mathbb{P}^2$  and  $\mathrm{dP}_7 = \mathrm{Bl}_2 \mathbb{P}^2$ . A *del Pezzo threefold* is a smooth projective threefold  $V$  with  $-K_V = 2H$  for an ample  $H$ ; its degree is  $H^3$ . For a polygon or polytope, edges are *unit* if they have lattice length one. Homology and cohomology have  $\mathbb{Q}$ -coefficients unless stated otherwise. The anticanonical cone over a del Pezzo surface  $S$  is  $C(S) = \mathrm{Spec} \bigoplus_{k \geq 0} H^0(S, -kK_S)$ .

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## 2. THE GERMS: LINKS, MINIVERSAL BASES, MILNOR FIBERS

Throughout this section  $(Y, 0) = C(S)$  is the anticanonical cone over a smooth del Pezzo surface  $S$ , an isolated Gorenstein toric threefold singularity when  $S$  is toric. For  $S = \mathrm{dP}_6$ ,  $\mathrm{dP}_7$  and  $\mathbb{F}_1$ , the cases used in this paper, it is the toric variety of the cone over the unit-edge reflexive hexagon, pentagon and quadrilateral respectively, placed at height one. (The unit-edge polygon here is the *fan* polygon of  $S$ ; its polar, the Newton polygon of  $-K_S$ , is not unit-edge for the pentagon.)

**2.1. Links.** The punctured cone is the total space of  $\mathcal{O}_S(K_S)$  minus its zero section, so the link  $L$  is the unit circle bundle of  $\mathcal{O}_S(K_S)$ , and the homology Gysin sequence together with  $H_1(S) = 0$  gives a canonical isomorphism

$$(1) \quad H_2(L; \mathbb{Q}) \cong K_S^\perp \subset H_2(S; \mathbb{Q}),$$

of rank 3 for  $\mathrm{dP}_6$  and 2 for  $\mathrm{dP}_7$ . For a node,  $L \cong S^2 \times S^3$  and  $H_2(L; \mathbb{Q}) = \mathbb{Q}$ . The lattice  $K^\perp(\mathrm{dP}_6)$  is the root lattice  $A_1 \oplus A_2$  (discriminant 6);  $K^\perp(\mathrm{dP}_7)$  has discriminant 7 and contains exactly one pair  $\pm$  of primitive classes of square  $-2$ .

**2.2. Miniversal deformations.** By Altman's theorems, the deformation theory of these germs is completely explicit.  $T^1$  is concentrated in the Gorenstein character degree, of dimension  $n - 3$  for an  $n$ -gon [Alt95, (6.5) and the subsequent Remark]; the versal family in that degree, constructed in [Alt97], is therefore a versal deformation of the affine cone, with base ring computed there:

$$\begin{aligned} \mathrm{dP}_7 &: \mathbb{C}[s_1, s_2]/(s_1^2, s_1 s_2) & [\text{Alt97}, (9.1)(\mathrm{v})], \\ \mathrm{dP}_6 &: \mathbb{C}[s_1, s_2, s_3]/(s_1 s_3, s_2 s_3) & [\text{Alt97}, (2.5)]. \end{aligned}$$

The reduced components correspond to the maximal decompositions of the polygon into Minkowski sums of lattice polygons [Alt97, (2.5), (8.1)]: the pentagon decomposes in exactly one way, into a unit segment (its pair of parallel edges) and a unimodular triangle, giving one reduced component, the line  $\{s_1 = 0\}$ ; the hexagon decomposes as a sum of two triangles (the line  $\{s_1 = s_2 = 0\}$ ) and as a sum of three segments (the plane  $\{s_3 = 0\}$ ).

We record three standard supplements used later. First, *minimality*: the Kodaira–Spencer map of Altman's family is an isomorphism onto  $T^1$  [Alt97, (6.2), (6.3)], so the versal family is miniversal. Second, *the germ*:  $T^1$  and  $T^2$  of the affine cone are finite-dimensional and supported at the vertex, and the deformation theory of the affine variety is identified with that of the analytic germ  $(Y, 0)$  by Artin approximation [Art69]; the analytic miniversal deformation exists by Grauert [Gra72]; the base is the analytic germ of the ring displayed above. (The versality criterion used in [Alt97, (7.2)] is that of de Jong–van Straten [dJvS94, §4].) Third, *arcs*: a morphism from the germ of a disk to the miniversal base factors through the reduction and, since the analytic local ring of a disk is a domain, lies in a single reduced component; for the pentagon base this forces  $s_1 \equiv 0$  along any arc, and for the hexagon base it forces the arc into the line or into the plane. We refer to the corresponding local smoothings as smoothings *through* the given component.

**2.3. Sweeps and Milnor fibers.** The smoothing components of these cones are cones swept by hyperplane sections, in the sense of Pinkham [Pin74, (4.2), (5.1), (5.5)]; see [PP15, Lemma 4.1, Proposition 4.2] for the statement in all dimensions in the pencil form used here. Concretely (Gross [Gro97, §5.5]; Petracci [Pet20, Proposition 2.2], [Pet19, Proposition 4.12]): the general fiber of the degree-7 smoothing is  $V_7 \setminus \bar{S}$  with  $V_7 = \mathrm{Bl}_{pt} \mathbb{P}^3$ ; the general fiber of the hexagon's one-parameter (line) component is  $(\mathbb{P}^1)^3 \setminus \bar{S}$ ; and that of its two-parameter (plane) component is  $W_6 \setminus \bar{S}$  with  $W_6 = \mathbb{P}(T_{\mathbb{P}^2})$  the divisor of bidegree  $(1, 1)$  in  $\mathbb{P}^2 \times \mathbb{P}^2$ . In each case  $V$  is a del Pezzo threefold of the same degree with  $\bar{S} \in |H|$  a copy of  $S$ , embedded by  $-K_S = H|_{\bar{S}}$ ; the classification of the possible  $V$  (Fujita [Fuj80, (5.8), (5.16)]) shows there are no others. For the record, the open Milnor fibers of the three branches have  $(b_2, b_3) = (1, 1)$ ,  $(2, 1)$  and  $(1, 2)$  respectively; the degree-6 values agree with [KP21, Proposition 3.5].

**Proposition 2.1.** *Let  $\mathcal{M} = V \setminus \bar{S}$  be one of the three sweep complements. Under the identification (1) of  $H_2$  of the link  $L = \partial(\text{tube around } \bar{S})$  with  $K_S^\perp$ ,*

$$\ker(H_2(L; \mathbb{Q}) \rightarrow H_2(\mathcal{M}; \mathbb{Q})) = \ker(K_S^\perp \rightarrow H_2(V; \mathbb{Q})) = \begin{cases} \text{the } (-2)\text{-line} & V = V_7, \\ A_1 & V = (\mathbb{P}^1)^3, \\ A_2 & V = W_6, \end{cases}$$

where the map is the restriction to  $K^\perp$  of the pushforward  $H_2(S; \mathbb{Q}) \rightarrow H_2(V; \mathbb{Q})$ . For a node,  $\ker(H_2(L; \mathbb{Q}) \rightarrow H_2(\mathcal{M}; \mathbb{Q})) = H_2(L; \mathbb{Q}) = \mathbb{Q}$ .

*Proof.* Since  $H_1(\bar{S}) = 0$ , the Thom isomorphism gives  $H_3(V, \mathcal{M}) \cong H_1(\bar{S}) = 0$ , so  $H_2(\mathcal{M}) \hookrightarrow H_2(V)$ , and the inclusion  $L \subset \mathcal{M}$  composed with this injection factors, up to homotopy

through the tube, as the projection  $L \rightarrow \bar{S}$  followed by  $\bar{S} \subset V$ ; the two kernels therefore coincide. (Here  $L$  is the circle bundle of the normal bundle  $\mathcal{O}_{\bar{S}}(\bar{S}) \cong \mathcal{O}_{\bar{S}}(-K_{\bar{S}})$ ; its Gysin sequence identifies  $H_2(L; \mathbb{Q})$  with the same subspace  $K^\perp$  as in (1).) The pushforward is computed in the standard bases. For  $V_7$ : a smooth member of  $|H| = |2H_{\mathbb{P}^3} - E|$  is the strict transform of a quadric through the blown-up point, so  $\bar{S} \cong \text{Bl}_2 \mathbb{P}^2$  with  $H_{\mathbb{P}^3}|_{\bar{S}} = 2\ell - e_1 - e_2$  and  $E|_{\bar{S}} = \ell - e_1 - e_2$ ; the pushforward matrix has rows  $(2, 1), (1, 1), (1, 1)$  on  $(\ell, e_1, e_2)$  and kernel  $\langle e_1 - e_2 \rangle$ , which is the  $(-2)$ -line. For  $(\mathbb{P}^1)^3$ : the three projections restrict to the conic pencils  $|\ell - e_i|$ , the rows are  $(1, 1, 1)$  and the unit vectors, and the kernel is  $\langle \ell - e_1 - e_2 - e_3 \rangle = A_1$ . For  $W_6$ : the two contractions restrict to  $\ell$  and  $2\ell - e_1 - e_2 - e_3$ , and the kernel is  $\langle e_1 - e_2, e_2 - e_3 \rangle = A_2$ . For the node,  $\mathcal{M} \simeq T^*S^3$  has  $H_2 = 0$ .  $\square$

*Remark 2.2.* The three kernels are the lattices  $\Xi(V)$  of Kuznetsov–Prokhorov: [KP23, Theorem 1.9(i)] states  $\text{Cl}(V) \cong \Xi(V)^\perp \subset \text{Cl}(\bar{S})$  with  $\Xi(V)$  negative definite of rank  $10 - d - \rho(V)$ , giving  $A_1, A_1, A_2$  for  $V_7, (\mathbb{P}^1)^3, W_6$ , an independent confirmation.

**2.4. Which sweep for which component.** The one-parameter (line) component of the hexagon base sweeps to  $(\mathbb{P}^1)^3$  and the two-parameter (plane) component to  $W_6$ ; this is stated in [Pet20, Proposition 2.2], follows from [Alt95, (7.1.5)] (the total spaces over the two components are the cones over  $(\mathbb{P}^1)^3$  and over  $\mathbb{P}^2 \times \mathbb{P}^2$ , and the hyperplane section of  $\mathbb{P}^2 \times \mathbb{P}^2$  in the Segre embedding is  $W_6$ ), and is confirmed numerically by the Betti numbers quoted above against [KP21, Proposition 3.5]. Theorem 6.5 uses only the nodes and the degree-7 branch. The matching is used in Lemma 3.3 for the degree-6 line branch, and through it in Theorem 6.1(3) and in the degree-6 assertion of Theorem 7.2.

### 3. VANISHING PERIODS

This section contains the analytic mechanism. Throughout,  $\mathcal{X} \rightarrow \mathbb{D}$  is a one-parameter smoothing of a connected compact complex threefold  $X$  with  $\omega_X \cong \mathcal{O}_X$ ,  $h^1(\mathcal{O}_X) = 0$  and isolated singularities of the types in Theorem A.

**3.1. A relative holomorphic 3-form.** Flatness and the Gorenstein property of all fibers make  $\mathcal{X} \rightarrow \mathbb{D}$  a Gorenstein morphism after shrinking the disk, so the relative dualizing sheaf  $\omega_{\mathcal{X}/\mathbb{D}}$  is invertible and compatible with base change [Con00] (in the analytic category this also follows from adjunction, as in Remark 4.5). We first prove  $c_1(\omega_{X_s}) = 0$  integrally. Choose the disjoint good representatives and exterior  $W$  of Section 4.1. Over a neighborhood of  $W$  the family is a smooth proper submersion with boundary and hence is  $C^\infty$ -trivial. Since  $\omega_X|_W \cong \mathcal{O}_W$ , the restriction of  $c_1(\omega_{\mathcal{X}/\mathbb{D}})$  to the corresponding exterior in  $X_s$  vanishes. On each local piece it also vanishes: choose first a Stein contractible neighborhood  $N_p$  of  $(p, 0)$  in  $\mathcal{X}$ , on which the invertible sheaf  $\omega_{\mathcal{X}/\mathbb{D}}$  is trivial, and then the good representative  $U_p$  inside  $N_p$ .

For every boundary link in this paper,  $H^1(L_p; \mathbb{Z}) = 0$ . This is immediate for the nodal link  $S^2 \times S^3$ . For a cone point,  $L_p$  is the circle bundle of  $\mathcal{O}_S(K_S)$  over the simply connected del Pezzo surface  $S$ ; the homotopy sequence shows that  $\pi_1(L_p)$  is the cyclic group  $\mathbb{Z}/d$ , where  $d$  is the divisibility of the Euler class  $K_S \neq 0$ , because the connecting map  $\pi_2(S) \rightarrow \pi_1(S^1) = \mathbb{Z}$  is evaluation against  $K_S$ . Hence  $H^1(L_p; \mathbb{Z}) = \text{Hom}(\mathbb{Z}/d, \mathbb{Z}) = 0$  for every del Pezzo cone point, whether or not  $K_S$  is primitive. (It is primitive for  $\text{dP}_6, \text{dP}_7$  and  $\mathbb{F}_1$ , where  $L_p$  is simply connected, but not for  $\mathbb{P}^1 \times \mathbb{P}^1$ .) The Mayer–Vietoris sequence for  $X_s = W \cup \bigsqcup_p \mathcal{M}_p$  therefore injects

$$H^2(X_s; \mathbb{Z}) \hookrightarrow H^2(W; \mathbb{Z}) \oplus \bigoplus_p H^2(\mathcal{M}_p; \mathbb{Z}).$$

The preceding local vanishings give  $c_1(\omega_{X_s}) = 0$ . Semicontinuity gives  $h^1(\mathcal{O}_{X_s}) = 0$ ; the exponential sequence then embeds  $\text{Pic}(X_s)$  into  $H^2(X_s; \mathbb{Z})$  and forces  $\omega_{X_s} \cong \mathcal{O}_{X_s}$ . (The hypothesis  $\omega_X \cong \mathcal{O}_X$  cannot be dropped: the cone over a smooth quadric surface in  $\mathbb{P}^4$  is a one-node threefold smoothed by the family of quadric threefolds, and the vanishing cycle of that smoothing is null-homologous; that example satisfies  $h^1(\mathcal{O}_X) = 0$  and is excluded here exactly because  $\omega_X \cong \mathcal{O}_X(-3)$  is nontrivial. The hypothesis  $h^1(\mathcal{O}_X) = 0$  enters only through the exponential sequence in the previous sentence.) Constants and upper semicontinuity give  $h^0(\mathcal{O}_{X_s}) = 1$  after shrinking, hence  $h^0(\omega_{X_s}) = 1$ . By Grauert's base-change theorem [Gra60],  $\pi_*\omega_{\mathcal{X}/\mathbb{D}}$  is then a line bundle on the shrunk disk; we fix a trivializing section  $\Omega$ , whose restriction  $\Omega_s$  generates the invertible sheaf  $\omega_{X_s}$  and is therefore a nowhere-vanishing holomorphic 3-form on the smooth locus of  $X_s$  for  $s \neq 0$  and a generator of  $\omega_X$  near each singular point at  $s = 0$ .

### 3.2. Nodes.

**Lemma 3.1.** *Let  $p$  be an ordinary double point of  $X$ . Then the vanishing-cycle class  $\delta_p(s) \in H_3(X_s; \mathbb{Q})$  is nonzero for all small  $s \neq 0$ .*

*Proof.* The germ  $(X, p)$  is  $\{xy - zw = 0\} \subset (\mathbb{C}^4, 0)$ , with miniversal deformation  $\{xy - zw = u\}$  over the  $u$ -line [KS72]. The family induces a classifying map  $s \mapsto \tilde{u}(s)$ , holomorphic with  $\tilde{u}(0) = 0$  (only its nonvanishing at each fixed  $s \neq 0$  is used, which is independent of choices), and  $X_s$  is smooth near  $p$  exactly when  $\tilde{u}(s) \neq 0$ ; the hypothesis gives  $\tilde{u}(s) \neq 0$  for all small  $s \neq 0$ . The pullback total space is the hypersurface  $\{xy - zw = \tilde{u}(s)\}$ , Gorenstein even at its singular points, and adjunction exhibits  $\omega_{\mathcal{X}/\mathbb{D}}$  near  $p$  as generated by the relative Gelfand–Leray form  $\text{GL}_u = \text{Res}[dx dy dz dw / (xy - zw - u)]$ . Hence  $\Omega = g \cdot \text{GL}$  with  $g$  holomorphic on the total-space germ and  $g(p, 0) \neq 0$ ; extend  $g$  to an ambient holomorphic function so that  $g(p, s)$  is defined.

After the linear change putting  $xy - zw$  into the form  $\sum_{i=1}^4 v_i^2$ , the vanishing cycle over  $u$  is the real slice  $\delta_u = \sqrt{u} \cdot S^3$ , whose class is single-valued: in fiber dimension three the Picard–Lefschetz transvection fixes  $\delta$  (as  $\langle \delta, \delta \rangle = 0$ ), and the antipodal ambiguity of  $\sqrt{u}$  acts on  $S^3$  with degree  $+1$ . Under  $v \mapsto \lambda v$  one has  $u \mapsto \lambda^2 u$  and  $\text{GL} \mapsto \lambda^2 \text{GL}$ , so the local period is exactly linear:

$$\int_{\delta_u} \text{GL}_u = c_0 u, \quad c_0 = 4 \int_{S^3} \frac{dv_1 dv_2 dv_3}{2v_4} = 4\pi^2 \neq 0$$

up to orientation conventions, the factor 4 being the Jacobian determinant of the linear change of coordinates. For the twisted form, expanding the ambient extension  $g = g(p, s) + O(|v|)$  and using  $|v| = |u|^{1/2}$  on  $\delta_u$  gives

$$F(u, s) := \int_{\delta_u} g \cdot \text{GL}_u = u (c_0 g(p, s) + O(|u|^{1/2})),$$

so  $F(u, s) = u \cdot (\text{unit})$  near  $(0, 0)$ ; here  $u, s$  are independent until the substitution  $u = \tilde{u}(s)$ .

Suppose  $\delta_p(s_0) = 0$  in  $H_3(X_{s_0}; \mathbb{Q})$  for some small  $s_0 \neq 0$ . A nonzero integer multiple of the cycle then bounds, and  $\Omega_{s_0}$  is closed, so  $\int_{\delta_p(s_0)} \Omega_{s_0} = 0$ . But this period equals  $F(\tilde{u}(s_0), s_0) = \tilde{u}(s_0) \cdot (\text{unit}) \neq 0$ , a contradiction.  $\square$

**3.3. Del Pezzo cone points.** For the cone germs the same strategy applies, with the local model family replaced by the sweep and the explicit period computed on the sweep complement. Both relevant complements admit a uniform toric description. Let  $V$  be  $(\mathbb{P}^1)^3$  or  $\text{Bl}_{pt} \mathbb{P}^3$  with its fan in  $N \cong \mathbb{Z}^3$ , and let  $m \in N^\vee$  be the character  $(1, 1, 1)$ , respectively

$(1, 1, -1)$  in the basis where the fan of  $\text{Bl}_{pt} \mathbb{P}^3$  has rays  $e_1, e_2, e_3$ ,  $r = -e_1 - e_2 - e_3$  and  $e_1 + e_2 + e_3$ . In both cases  $\langle m, v \rangle = \pm 1$  on every ray, the closure  $\bar{S}$  of  $\{\chi^m = 1\}$  is a copy of the appropriate del Pezzo surface with  $[\bar{S}] = -\frac{1}{2}K_V$ , and

$$(2) \quad \Omega_{\mathcal{M}} = \frac{w}{(w-1)^2} \frac{dx}{x} \wedge \frac{dy}{y} \wedge \frac{dz}{z}, \quad w = \chi^m,$$

is the generator of  $\omega(V \setminus \bar{S}) \cong \mathcal{O}$ , unique up to scale: the order of (2) along a boundary divisor  $D_v$  is  $\langle m, v \rangle - 2 \min(\langle m, v \rangle, 0) - 1 = 0$ , and along  $\bar{S}$  it is  $-2$ . Here uniqueness is algebraic. If  $f \in \mathcal{O}(V \setminus \bar{S})^\times$ , then  $\text{div}_V(f) = n\bar{S}$  for some  $n \in \mathbb{Z}$ ; since  $\text{Pic}(V)$  is torsion-free and  $[\bar{S}] \neq 0$ , one has  $n = 0$ , and a rational function without zeros or poles on the projective variety  $V$  is constant.

Every sweep pair  $(V, \bar{S})$  of Section 2.3 with  $V = \text{Bl}_{pt} \mathbb{P}^3$  or  $V = (\mathbb{P}^1)^3$  is isomorphic to the toric pair just described. On  $\text{Bl}_{pt} \mathbb{P}^3$  the smooth members of  $|H| = |2H_{\mathbb{P}^3} - E|$  are the strict transforms of the smooth quadrics through the blown-up point. The group  $\text{PGL}_4$  acts transitively on smooth quadrics, and the automorphism group of a smooth quadric acts transitively on its points, so the automorphisms of  $\mathbb{P}^3$  fixing the point act transitively on these members. On  $(\mathbb{P}^1)^3$ , with homogeneous coordinates  $[x_0 : x_1]$ ,  $[y_0 : y_1]$ ,  $[z_0 : z_1]$  on the three factors (so that  $x = x_1/x_0$ ,  $y = y_1/y_0$ ,  $z = z_1/z_0$  above), a member of  $|H|$  is the zero locus of a trilinear form  $z_0 A_0(x, y) + z_1 A_1(x, y)$  with  $A_0, A_1$  bilinear. It is smooth exactly when the binary quadratic form  $\det(\lambda A_0 + \mu A_1)$  has nonzero discriminant, Cayley's hyperdeterminant [GKZ94, Chapter 14, §§1 and 4]. A linear change of  $(z_0, z_1)$  then moves the two roots to  $\lambda = 0$  and  $\mu = 0$ , so that  $A_0$  and  $A_1$  have rank one with invertible sum, and a linear change of  $x$  and of  $y$  brings the form to  $x_0 y_0 z_0 - x_1 y_1 z_1$ , the equation of the closure of  $\{\chi^m = 1\}$ .

**Lemma 3.2.** *In each of the two complements  $\mathcal{M} = V \setminus \bar{S}$  above there is a closed 3-cycle  $T \subset \mathcal{M}$  with  $\int_T \Omega_{\mathcal{M}} = \pm 4\pi^2$ . Consequently  $[\Omega_{\mathcal{M}}] \neq 0$  in  $H^3(\mathcal{M}; \mathbb{C})$ , and since  $b_3(\mathcal{M}) = 1$ , the period pairing  $H_3(\mathcal{M}; \mathbb{Q}) \rightarrow \mathbb{C}$ ,  $\nu \mapsto \int_\nu \Omega_{\mathcal{M}}$ , is injective.*

*Proof.* Choose an edge  $\langle v_+, v_- \rangle$  of the fan with  $\langle m, v_\pm \rangle = \pm 1$  and an adjacent maximal cone  $\langle v_+, v_-, n_0 \rangle$  with  $\langle m, n_0 \rangle = +1$ , such that the maximal cone  $\langle v_+, v_-, n_1 \rangle$  on the other side of the cone  $\langle v_+, v_- \rangle$  has  $\langle m, n_1 \rangle = -1$ . This last condition keeps the cap  $T_3$  below off  $\bar{S}$ ; without it the cap can meet  $\bar{S}$ . The two cases are

| $V$                           | $(v_+, v_-, n_0; n_1)$   | transition to the $n_1$ chart | $c$ |
|-------------------------------|--------------------------|-------------------------------|-----|
| $(\mathbb{P}^1)^3$            | $(e_1, -e_2, e_3; -e_3)$ | $(p, q, 1/z)$                 | 0   |
| $\text{Bl}_{pt} \mathbb{P}^3$ | $(e_1, r, e_2; e_3)$     | $(p/z, q/z, 1/z)$             | -1  |

where  $c$  is the  $z$ -exponent of the transition monomial for  $p'$ . In the unimodular chart with coordinates  $(p, q, z)$  dual to  $(v_+, v_-, n_0)$  one has  $w = pq^{-1}z$ ,  $\bar{S} = \{t = 0\}$  with  $t = q - pz$ , and  $\Omega_{\mathcal{M}} = \pm dp \wedge dq \wedge dz/t^2$ . Fix  $\varepsilon > 0$ ,  $R > 1$  and set

$$T_1 = \{p = 0, q = \varepsilon R e^{i\psi}, |z| \leq R\},$$

$$T_2 = \{p = r\varepsilon e^{i\theta}, q = -(1-r)\varepsilon R e^{i(\theta+\varphi)}, z = R e^{i\varphi}\}, \quad r \in [0, 1],$$

$$T_3 = \{p' = \varepsilon R^c e^{i\psi}, q' = 0, |z'| \leq 1/R\} \quad \text{in the chart at the opposite corner } n_1,$$

where  $(p', q', z')$  are the dual coordinates of  $(v_+, v_-, n_1)$  and  $c$  is as in the table. The corner-chart cap is necessary: in the  $\text{Bl}_{pt} \mathbb{P}^3$  case the naive closure of  $\{|p| = \varepsilon, q = 0, |z| \geq R\}$  collapses onto the torus-fixed point of the corner chart, which lies on  $\bar{S} = \{p' = q'z'\}$ ; the cap  $T_3$  instead keeps  $|p' - q'z'| = \varepsilon R^c > 0$ . On  $T_2$ ,  $t = -\varepsilon R e^{i(\theta+\varphi)}$  has constant modulus;

the  $r = 0$  boundary is the boundary torus of  $T_1$  and the  $r = 1$  boundary maps under the transition onto  $\partial T_3$ , so with orientations of the caps chosen to cancel  $\partial T_2$ ,  $T = T_1 + T_2 + T_3$  is a closed 3-cycle in  $\mathcal{M}$ .

The pullback of  $\Omega_{\mathcal{M}}$  to  $T_1$  and  $T_3$  vanishes identically ( $dp \mapsto 0$ , respectively  $dq' \mapsto 0$ , while  $\Omega_{\mathcal{M}}$  is regular there), and on  $T_2$  the pullback of  $dp \wedge dq \wedge dz/t^2$  is the *constant* form  $dr \wedge d\theta \wedge d\varphi$ : writing it out, the complex Jacobian of the parametrization equals  $t^2$  exactly. Hence  $\int_T \Omega_{\mathcal{M}} = \int_0^1 \int_0^{2\pi} \int_0^{2\pi} dr d\theta d\varphi = 4\pi^2$  up to the orientation sign.

Finally  $b_3(\mathcal{M}) = 1$ : from the pair sequence of  $(V, \mathcal{M})$  and  $H^3(V) = 0$ ,  $b_3(\mathcal{M})$  is the corank of the restriction  $H^2(V; \mathbb{Q}) \rightarrow H^2(\bar{S}; \mathbb{Q})$ , which is one in both cases.  $\square$

**Lemma 3.3.** *Let  $p$  be a singular point of  $X$  whose germ is the anticanonical cone over  $\mathrm{dP}_7$ , or over  $\mathrm{dP}_6$  with the induced local smoothing through the one-parameter component. Then  $H_3(\mathcal{M}_p; \mathbb{Q}) (\cong \mathbb{Q}) \rightarrow H_3(X_s; \mathbb{Q})$  is injective for all small  $s \neq 0$ .*

*Proof.* By Section 2.2 the induced arc lies in a single reduced component  $c$  of the miniversal base of the germ, off the local discriminant for  $s \neq 0$ , and by Section 2.3 the Milnor fiber is the corresponding sweep complement, which for the two branches in the statement is isomorphic to one of the complements of Lemma 3.2 by the paragraph preceding that lemma. The component carries Pinkham's  $\mathbb{C}^*$ -equivariant swept family with parameters of weight one (Section 2.2:  $T^1$  sits in the Gorenstein degree); let  $\Omega^{\mathrm{loc}}$  be an equivariant algebraic generator of its relative dualizing sheaf, homogeneous of some integral weight  $a$ . To construct it, work in the graded local ring at the vertex. The relative dualizing module is invertible, and graded Nakayama selects from any algebraic local generator a homogeneous component that still generates. Call  $U$  the open locus on which that homogeneous generator generates;  $U$  is  $\mathbb{C}^*$ -stable, being the nonvanishing locus of a homogeneous section, and it contains the vertex. Then  $U$  contains the whole fiber  $b = 1$ . Let  $x$  be a point of that fiber. Scaling the positive-weight cone coordinates and the weight-one parameter to zero carries  $x$  to the vertex, so the vertex lies in the closure of  $\mathbb{C}^* \cdot x$ ; since  $U$  is open and contains the vertex, some  $\lambda \cdot x$  lies in  $U$ , and since  $U$  is  $\mathbb{C}^*$ -stable,  $x$  does. Thus the restriction to  $b = 1$  is an algebraic generator of  $\omega_{\mathcal{M}}$ , hence a constant multiple of  $\Omega_{\mathcal{M}}$  by the unit calculation following (2); rescale it so that  $\Omega^{\mathrm{loc}}|_{b=1} = \Omega_{\mathcal{M}}$ . For a horizontal family  $\nu$  of classes in  $H_3(\mathcal{M}; \mathbb{Q})$ ,

$$\int_{\nu(b)} \Omega_b^{\mathrm{loc}} = c(\nu) \cdot \chi(b),$$

with  $\chi$  the weight- $a$  monomial in the component parameter and  $c(\nu) = \int_{\nu} \Omega_{\mathcal{M}}$ : single-valuedness holds because the  $\mathbb{C}^*$ -action is a group action on the total space with parameter weight one, so transport once around the base circle is the identity; and  $c(\cdot)$  is linear. Writing  $\Omega = g \cdot \Omega^{\mathrm{loc}}$  near  $p$  with  $g$  a unit as in Section 3.1, let  $b = b(s)$  be the actual parameter of the component. The cycle obtained from the fixed compact cycle of Lemma 3.2 by the  $\mathbb{C}^*$ -action lies in the local Milnor piece and converges uniformly to  $p$ , because all cone coordinates have positive weights. Hence  $g|_{\nu(s)} = g(p, 0) + o(1)$  uniformly, while homogeneity factors the same monomial  $\chi(b(s))$  from the main term and the error. The global period is therefore

$$\int_{\nu(s)} \Omega_s = \chi(b(s)) (c(\nu) g(p, 0) + o(1)).$$

If  $\nu \neq 0$  in  $H_3(\mathcal{M}; \mathbb{Q})$  had  $\nu_* = 0$  in  $H_3(X_s; \mathbb{Q})$  for arbitrarily small  $s \neq 0$ , then a rational multiple of  $\nu(s)$  would bound and the closed form  $\Omega_s$  would have zero period, while  $\chi(b(s)) \neq 0$  off the discriminant; letting  $s \rightarrow 0$  along such values gives  $c(\nu) g(p, 0) = 0$  and, since  $g(p, 0) \neq 0$ ,  $c(\nu) = 0$ , contradicting the injectivity of the period pairing in Lemma 3.2.

(That  $H_3(\mathcal{M}; \mathbb{Q})$  has rank one is what converts this class-by-class exclusion into injectivity for all small  $s$ ; a higher-rank branch would require a uniform argument.)  $\square$

#### 4. SURGERY MEMBERSHIP

**4.1. The decomposition.** Let  $p_1, \dots, p_r$  be the singular points of  $X$ , with disjoint closed cone neighborhoods  $U_p$  chosen inside Milnor balls, and  $W = X \setminus \bigcup_p \text{int } U_p$ , so  $\partial W = \bigsqcup_p L_p$ . Good representatives exist for arbitrary isolated singularities (Greuel [Gre20, Definition 2.12, Lemma 2.13, Theorem 2.15]), and for small  $s$  the family is smooth, hence  $C^\infty$ -trivial, over a neighborhood of  $W$ ; therefore

$$X_s \cong W \cup_{\bigsqcup L_p} \bigsqcup_p \mathcal{M}_p, \quad \widehat{X} \cong W \cup_{\bigsqcup L_p} \bigsqcup_p N_p,$$

where  $\mathcal{M}_p$  is the Milnor fiber of the induced local smoothing and  $N_p$  a neighborhood of the exceptional set of the fixed crepant resolution. Both gluings are along the same links. Only the  $C^\infty$ -triviality over a neighborhood of  $W$  is used here, so the same decomposition of  $X_s$  along  $\bigsqcup_p L_p$  holds for an arbitrary one-parameter deformation, with  $X_s \cap U_p$  in place of  $\mathcal{M}_p$  at any point whose germ is not smoothed; this is the form Section 4.3 uses.

**Lemma 4.1** (membership). *For every  $\gamma \in H_3(X_s; \mathbb{Q})$ , the Mayer–Vietoris boundary  $\partial\gamma \in \bigoplus_p H_2(L_p; \mathbb{Q})$  lies in the space  $K(X)$  of Theorem A: its components die in every  $H_2(\mathcal{M}_p; \mathbb{Q})$ , and their sum dies in  $H_2(W; \mathbb{Q})$ , hence in  $H_2(\widehat{X}; \mathbb{Q})$ . Moreover, at a node  $q$  the  $L_q$ -component of  $\partial\gamma$  is a fixed nonzero multiple of  $(\gamma \cdot \delta_q)$  times the generator, and at a cone point  $p$  the boundary map identifies  $H_3(\mathcal{M}_p, L_p; \mathbb{Q}) \xrightarrow{\sim} \mathcal{R}_p$ .*

*Proof.* The first statement is Mayer–Vietoris exactness for  $X_s = W \cup \bigsqcup \mathcal{M}_p$ , together with  $W \subset \widehat{X}$ . At a node,  $\mathcal{M} \simeq T^*S^3$ :  $H_3(\mathcal{M}, L; \mathbb{Q}) \cong H^3(\mathcal{M}; \mathbb{Q}) = \mathbb{Q}$  is generated by a cotangent fiber  $F$  with  $\partial F$  the  $S^2$ -generator of  $H_2(S^2 \times S^3)$  and  $F \cdot \delta = 1$ ; excision identifies the  $L_q$ -component of  $\partial\gamma$  with  $\partial$  of the image of  $\gamma$  in  $H_3(\mathcal{M}_q, L_q)$ , which equals  $(\gamma \cdot \delta_q)F$  by the perfect pairing. At a cone point, exactness of the pair sequence makes the image of  $\partial: H_3(\mathcal{M}, L; \mathbb{Q}) \rightarrow H_2(L; \mathbb{Q})$  equal to  $\ker(H_2(L; \mathbb{Q}) \rightarrow H_2(\mathcal{M}; \mathbb{Q})) = \mathcal{R}_p$ , while Lefschetz duality gives  $\dim H_3(\mathcal{M}, L; \mathbb{Q}) = b_3(\mathcal{M})$ , which by Section 2.3 equals 1, 1, 2 in the three branches, the rank of  $\mathcal{R}_p$  computed in Proposition 2.1. (The comparison of the actual Milnor fiber with the sweep complement as a manifold with boundary, which this uses, is Lemma 4.2 below. When  $b_3(\mathcal{M}) = 1$ , injectivity of  $\partial$  needs no comparison: the map  $H_3(\mathcal{M}) \rightarrow H_3(\mathcal{M}, L)$  is Lefschetz dual to a skew form on a line, hence zero.) A surjection between spaces of equal finite dimension is an isomorphism, so  $\partial$  is injective with image  $\mathcal{R}_p$ ; equivalently  $H_3(\mathcal{M}) \rightarrow H_3(\mathcal{M}, L)$  is zero,  $H_3(L) \rightarrow H_3(\mathcal{M})$  is surjective, and the skew intersection form of  $H_3(\mathcal{M}; \mathbb{Q})$  vanishes.  $\square$

**4.2. Boundary compatibility.** Lemma 4.1 uses the subspaces  $\mathcal{R}_p \subset H_2(L_p)$  through the *sweep-side* description of Proposition 2.1 (the unit normal circle bundle of  $\bar{S} \subset V$ ), while the map to  $H_2(\widehat{X})$  and all coordinates in Section 5 use the *resolution-side* description (the projection  $L_p \rightarrow E_p$  of the cone link onto the exceptional del Pezzo, the circle-bundle projection recalled after Lemma 4.2). These are a priori different identifications of  $H_2(L_p; \mathbb{Q})$  with  $K^\perp$ , and an arbitrary boundary diffeomorphism need not preserve either marking. Their compatibility is therefore a statement, not a formality. It appears not to be available in the literature: the identification of the germ Milnor fiber with the sweep complement is standard as *open* manifolds ([PP15, Proposition 4.2]), but the manifold-with-boundary form, compatible with the canonical link identification, is what the following lemma supplies.

**Lemma 4.2** (boundary compatibility). *Let  $(X, p)$  be analytically isomorphic to the anticanonical cone over the del Pezzo surface  $S$ , smoothed by a one-parameter family through the component  $c$  with sweep  $V \supset \bar{S}$ . Then the boundary identification  $\partial\mathcal{M}_p \cong L_p$  arising from the surgery decomposition is such that the sweep-side projection  $H_2(L_p; \mathbb{Q}) \rightarrow H_2(\bar{S}; \mathbb{Q})$  and the circle-bundle projection  $H_2(L_p; \mathbb{Q}) \rightarrow H_2(E_p; \mathbb{Q})$  agree after composing the former with the isometry  $H_2(\bar{S}; \mathbb{Z}) \rightarrow H_2(E_p; \mathbb{Z})$  induced by an isomorphism  $\bar{S} \cong E_p$ , which carries  $K_{\bar{S}}$  to  $K_{E_p}$ . Consequently  $\mathcal{R}_p$ , read in resolution-side coordinates, is the root subspace of Proposition 2.1.*

Here the germ-local crepant resolution of  $C(S)$  is the blowup of the vertex, its exceptional divisor is the zero section  $E_p \cong S$  of the total space of  $\mathcal{O}_S(K_S)$ , the link  $L_p$  is the unit circle bundle of that line bundle, and the *circle-bundle projection* is the bundle projection  $L_p \rightarrow E_p$ ; this is the identification used in (1). This local form is part of the standing hypotheses and is not automatic for a crepant resolution: a triangulation of a pentagon or hexagon that cuts off a unimodular corner gives a toric crepant resolution with a smaller toric surface over  $p$  together with an exceptional curve.

*Proof.* Let  $\mathcal{Y}_c \rightarrow B_c$  be the  $\mathbb{C}^*$ -equivariant Altmann–Pinkham family on the reduced component  $c$ . All component parameters have weight one and all affine cone coordinates have positive weights, so an equivariant closed embedding  $\mathcal{Y}_c \hookrightarrow \mathbb{A}^N \times B_c$  with fiber coordinate weights  $w_1, \dots, w_k > 0$  exists, and we let  $\bar{\mathcal{Y}}_c \rightarrow B_c$  be the fiberwise closure of  $\mathcal{Y}_c$  in  $\mathbb{P}(1, w_1, \dots, w_k) \times B_c$ . (For the del Pezzo surfaces of degrees 6 and 7 the anticanonical ring is generated in degree one, so all  $w_i$  may be taken equal to 1 and the ambient space is  $\mathbb{P}^k \times B_c$ ; general weights are kept only for uniformity of notation.) This is Pinkham’s sweeping out of the cone [Pin74, (4.2), (5.1)] carried out in families over  $B_c$ : the divisor at infinity is cut by the homogenizing coordinate, on which  $\mathbb{C}^*$  acts trivially, and is Proj of the graded coordinate ring of the cone, hence the fixed family  $\bar{S} \times B_c$ , independent of  $b$  because the parameters have positive weight. For  $b$  off the discriminant the pair  $(\bar{\mathcal{Y}}_b, \bar{S})$  is the polarized sweep pair  $(V, \bar{S})$  of Proposition 2.1, which is where its smoothness comes from. Choose an equivariant affine embedding, write  $w_1, \dots, w_k > 0$  for the coordinate weights and, for a common multiple  $d$  of them, use the weighted radius

$$\rho(z) = \left( \sum_{i=1}^k |z_i|^{2d/w_i} \right)^{1/(2d)}, \quad \rho(r \cdot z) = r\rho(z) \quad (r > 0).$$

Fix a sufficiently small weighted Milnor radius  $\varepsilon$  and a small sphere of radius  $\eta$  in  $B_c$ . Every sufficiently small parameter  $\beta \in c$  off the discriminant is uniquely  $\beta = tb$  with  $t > 0$  and  $\|b\| = \eta$ . The discriminant is homogeneous, so  $b$  is also off it. The action by  $t^{-1}$  gives a diffeomorphism of pairs

$$(3) \quad (\mathcal{Y}_\beta \cap \{\rho \leq \varepsilon\}, \mathcal{Y}_\beta \cap \{\rho = \varepsilon\}) \cong (K_{t,b}, \Sigma_{t,b}), \quad K_{t,b} := \mathcal{Y}_b \cap \{\rho \leq \varepsilon/t\}.$$

The pair on the left is the actual compact Milnor fiber and its boundary, not merely its interior.

We next identify the pair on the right with the sweep compact core. Near the fixed divisor  $S \subset \bar{\mathcal{Y}}_b$ , choose a normal coordinate  $q$ , and pass to the real oriented blowup along  $S$ . An affine coordinate of weight  $w_i$  has the form

$$z_i = q^{-w_i} (a_i(y, b) + O(q)),$$

where  $y \in S$  and the leading coefficients do not vanish simultaneously. Consequently

$$(4) \quad \rho = |q|^{-1} A(y, \arg q, |q|, b)$$

with  $A$  smooth and strictly positive on the oriented boundary. For  $t$  small, the equation  $\rho = \varepsilon/t$  therefore has a unique solution for  $|q|$  and is a smooth radial graph over the unit normal circle bundle of  $S$ . Normal radial flow identifies this graph with the boundary of a fixed tube of  $S$ , preserving the projection to  $S$ , and extends across the intervening collar. Thus (3) continues to a diffeomorphism of pairs

$$(\mathcal{M}_p, \partial\mathcal{M}_p) \cong (\overline{\mathcal{Y}}_b \setminus \text{int Tube}(S), \partial\text{Tube}(S))$$

that carries the actual boundary inclusion to the tube-boundary inclusion.

By the sweep description of Section 2.3, the compactified smooth pair  $(\overline{\mathcal{Y}}_b, S)$  is the corresponding polarized sweep pair  $(V, \bar{S})$ ; along a connected smooth branch the pair identifications may be transported by Ehresmann, using a trivializing vector field that is tangent to the fixed family  $S \times B_c$  and horizontal along it, so that  $S$  is fixed pointwise.

It remains to identify the marking of that boundary. Apply the same construction to the radial family  $\mathcal{Y}_{\tau b}$  for  $0 < \tau \leq t$ . On the real oriented blowup write  $r = |q|$  and  $\theta = \arg q$ . By (4) the boundary of  $K_{\tau, b}$  is  $\{r = (\tau/\varepsilon)A(y, \theta, r, b)\}$ , and the implicit function theorem gives a unique solution  $r^*(\tau, y, \theta) = (\tau/\varepsilon)A(y, \theta, 0, b) + O(\tau^2)$ , smooth up to  $\tau = 0$ . Let  $\Psi_\tau(y, \theta) \in \mathcal{Y}_{\tau b} \cap \{\rho = \varepsilon\}$  be the image under the action of  $\tau$  of the point with coordinates  $(y, \theta, r^*)$ . Its affine coordinates are  $(\tau/r^*)^{w_i} e^{-i\theta w_i} (a_i(y, b) + O(r^*))$ , which converge smoothly as  $\tau \rightarrow 0$  to a point of the link of the central cone. Thus  $\Psi$  trivializes the boundary bundle over the segment  $[0, t] \cdot b$ , and for every  $\tau$  the tube projection composed with  $\Psi_\tau$  is  $(y, \theta) \mapsto y$ . At  $\tau = 0$  the projection sends a nonzero point of the central cone to its projective class in  $S$ , while the normal circle is its  $S^1$ -orbit. It is therefore exactly the circle-bundle projection  $L_p \rightarrow S$  of the cone link.

It remains to compare this radial transport with the identification  $\partial\mathcal{M}_p \cong L_p$  that Section 4.1 fixes, which is not a free choice. The two live over different parameter spaces, so name the map between them: let  $\beta: (\mathbb{D}, 0) \rightarrow (B_c, 0)$  classify the induced local deformation, so that after shrinking the local family is  $\beta^*\mathcal{Y}_c$ , and take the good representative  $U_p$  of Section 4.1 with  $\partial U_p$  corresponding to  $\{\rho = \varepsilon\}$ . The comparison is in three steps: the boundary is a locally trivial bundle over the whole base, discriminant included; two trivializations of it over the disc induce the same map on  $H_2$ ; and the radial transport is one such trivialization.

First,  $\{\rho = \varepsilon\} \cap \mathcal{Y}_b$ , for  $b \in B_c$ , is a locally trivial  $C^\infty$  bundle over all of  $B_c$ , discriminant included. The relative singular locus is proper over the compact ball  $\{\rho \leq \varepsilon\}$  and meets the central fiber only at the vertex, so for small  $b$  it stays inside  $\{\rho < \varepsilon\}$ ; the total space is smooth near  $\{\rho = \varepsilon\}$ ; and  $\mathcal{Y}_b$  meets  $\{\rho = \varepsilon\}$  transversally, which holds at  $b = 0$  by  $\rho(r \cdot z) = r\rho(z)$  and is open in  $b$  by compactness. Ehresmann applies.

Second, fix a trivialization  $T$  of that bundle. Then  $\beta^*T$  and the trivialization of Section 4.1 are two trivializations of the same bundle over  $\mathbb{D}$ . Two such identify a fiber with the fiber over 0 by maps differing by  $g(0)^{-1}g(s)$  for some  $g$  into the diffeomorphism group of  $L_p$ ; a path from 0 to  $s$  makes  $t \mapsto g(\gamma(t))$  an isotopy from  $g(0)$  to  $g(s)$ , so the discrepancy is isotopic to the identity and acts trivially on  $H_2$ . Only path-connectedness is used.

Third, the radial transport trivializes the same bundle over the ray through  $\beta(s)$ , so on that ray it agrees with  $T$  on  $H_2$  for the same reason. The sweep-side tube projection and the resolution-side circle-bundle projection therefore induce the same map on  $H_2$ , up to the isometry of the statement.

Finally the  $A_1$  and  $A_2$  summands for  $\text{dP}_6$  and the unique  $(-2)$ -line for  $\text{dP}_7$  are intrinsic to the polarized lattice, so they correspond. Since the displayed diffeomorphism is one of pairs, it computes the kernel of the actual inclusion  $H_2(\partial\mathcal{M}_p) \rightarrow H_2(\mathcal{M}_p)$ , as required.  $\square$

**4.3. Deformations that leave germs singular.** Theorem A concerns smoothings. A threefold carrying a germ with no smoothing at all admits none, but it may still admit deformations that smooth its other germs, and the surgery argument constrains those as well. We record the form of Lemma 4.1 that Section 7 uses. Call an isolated singularity *reduced-rigid* if the reduction of its miniversal base is a single point; this does not assert  $T^1 = 0$ . The anticanonical cone over  $\mathbb{F}_1 = \text{Bl}_1 \mathbb{P}^2$ , one of the two del Pezzo surfaces of degree 8 (not  $\mathbb{P}^1 \times \mathbb{P}^1$ , whose cone is smoothable), is reduced-rigid: its quadrilateral, with edge vectors  $(1, 0), (1, 2), (-2, -1), (0, -1)$ , admits no nontrivial Minkowski decomposition *into lattice polygons*, although its cone of rational Minkowski summands is two-dimensional, so that  $\dim T^1 = 1$  and the miniversal base is  $\text{Spec } \mathbb{C}[\varepsilon]/(\varepsilon^2)$  [Alt97, (9.1)(iv), (9.2)]; see also Gross [Gro97, §5.5] and the example of [Gro97, §2] (Example 2.8 of the 1995 preprint, Example 2.19 of the 2025 revision arXiv:alg-geom/9506022v2), and [Wue26a].

**Lemma 4.3.** *Let  $(Y, 0)$  be reduced-rigid and let  $\mathcal{X} \rightarrow \mathbb{D}$  be a one-parameter deformation of a complex space  $X$  with  $(X, p) \cong (Y, 0)$ . Then the induced deformation of the germ at  $p$  is trivial. Consequently there are a neighborhood  $U$  of  $p$  in  $\mathcal{X}$  and an  $\varepsilon > 0$  with  $U \cong U_0 \times \mathbb{D}_\varepsilon$  over  $\mathbb{D}_\varepsilon$ , where  $U_0$  is a neighborhood of  $p$  in  $X$ , so that for  $0 < |s| < \varepsilon$  the fiber  $X_s \cap U$  has a singular point  $q_s$ , with  $q_s \rightarrow p$  and  $(X_s, q_s) \cong (Y, 0)$ .*

*Proof.* Near  $p$  the family is induced from the miniversal deformation by a holomorphic map  $c: (\mathbb{D}, 0) \rightarrow (B, 0)$ . The analytic local ring of a disk is a domain, hence reduced, so  $c$  factors through  $B_{\text{red}}$ , a reduced point; thus  $c$  is constant, and the pullback of the miniversal family along a constant map is the product  $(Y, 0) \times (\mathbb{D}, 0)$ . The final statement is the passage from the germ at  $(p, 0)$  to fibers, by choosing representatives of that product decomposition.  $\square$

**Lemma 4.4** (membership, with germs left singular). *Let  $X$  and  $\widehat{X}$  be as in the setting fixed in Section 1 before Theorem A, except that  $X$  may carry in addition finitely many anticanonical cones over  $\mathbb{F}_1$ , with  $\widehat{X} \rightarrow X$  the blowup of the vertex, with exceptional surface  $\mathbb{F}_1$ , at each of them; no smoothing of  $X$  is assumed to exist. Let  $\mathcal{X} \rightarrow \mathbb{D}$  be any one-parameter deformation of  $X$  and let*

$$\Sigma = \{p \in \text{Sing } X : \text{the local fiber } X_s \cap U_p \text{ is singular}\},$$

*a set independent of  $s$  for small  $s \neq 0$ , because each local classifying arc either lies in the local discriminant or meets it only at the origin. Put*

$$X_s^b = W \cup \bigsqcup_{p \notin \Sigma} \mathcal{M}_p,$$

*a compact oriented 6-manifold with boundary  $\bigsqcup_{p \in \Sigma} L_p$  contained in the smooth locus of  $X_s$ . Then for every  $\gamma \in H_3(X_s^b, \partial X_s^b; \mathbb{Q})$  there is a class  $\kappa \in \bigoplus_p H_2(L_p; \mathbb{Q})$  with*

$$\kappa \in \ker \left( \bigoplus_p H_2(L_p; \mathbb{Q}) \rightarrow H_2(\widehat{X}; \mathbb{Q}) \right), \quad \kappa_p \in \mathcal{R}_p \quad (p \notin \Sigma \text{ a cone point}),$$

*whose component at a node  $q \notin \Sigma$  is a fixed nonzero multiple of  $(\gamma \cdot \delta_q)$  times the generator, and whose component at a cone point  $p \notin \Sigma$  is the image of  $\gamma$  under the composite  $H_3(X_s^b, \partial X_s^b; \mathbb{Q}) \rightarrow H_3(\mathcal{M}_p, L_p; \mathbb{Q}) \xrightarrow{\partial, \sim} \mathcal{R}_p$ . Here  $\gamma \cdot \delta_q$  is the Poincaré–Lefschetz pairing of  $X_s^b$ . No local condition is imposed on  $\kappa_p$  for  $p \in \Sigma$ ; those coordinates are of course still coupled to the others by the displayed kernel condition.*

*Proof.* By Lemma 4.3 every  $\mathbb{F}_1$ -cone point lies in  $\Sigma$ , so a cone point  $p \notin \Sigma$  is a dP<sub>6</sub>- or dP<sub>7</sub>-cone point and the local computations of Lemma 4.1 apply to it. Apply Mayer–Vietoris

for the pair  $(X_s^b, \partial X_s^b)$  with  $A = W$ ,  $B = \bigsqcup_{p \notin \Sigma} \mathcal{M}_p$  and  $\partial X_s^b \subset A$ , the couple being made excisive by the standard collar thickening:

$$H_3(X_s^b, \partial X_s^b; \mathbb{Q}) \xrightarrow{\partial} \bigoplus_{p \notin \Sigma} H_2(L_p; \mathbb{Q}) \longrightarrow H_2(W, \partial X_s^b; \mathbb{Q}) \oplus \bigoplus_{p \notin \Sigma} H_2(\mathcal{M}_p; \mathbb{Q}).$$

The second summand gives  $(\partial\gamma)_p \in \mathcal{R}_p$  at the cone points  $p \notin \Sigma$ . For the first, let  $x \in H_2(W; \mathbb{Q})$  be the image of  $\partial\gamma$ . It dies in  $H_2(W, \partial X_s^b; \mathbb{Q})$ , so by exactness of

$$H_2(\partial X_s^b; \mathbb{Q}) \longrightarrow H_2(W; \mathbb{Q}) \longrightarrow H_2(W, \partial X_s^b; \mathbb{Q})$$

it is the image of some  $y \in H_2(\partial X_s^b; \mathbb{Q}) = \bigoplus_{p \in \Sigma} H_2(L_p; \mathbb{Q})$ . Set  $\kappa_p = (\partial\gamma)_p$  for  $p \notin \Sigma$  and  $\kappa_p = -y_p$  for  $p \in \Sigma$ . Then  $\kappa$  dies in  $H_2(W; \mathbb{Q})$ , hence in  $H_2(\widehat{X}; \mathbb{Q})$  because  $W \subset \widehat{X}$ .

For the local statements, fix a node  $q \notin \Sigma$ . Since  $\partial X_s^b \subset W$  is disjoint from  $\mathcal{M}_q$ , excision gives a restriction  $H_3(X_s^b, \partial X_s^b; \mathbb{Q}) \rightarrow H_3(\mathcal{M}_q, L_q; \mathbb{Q})$ , and under Poincaré–Lefschetz duality on  $X_s^b$  it is dual to the inclusion  $H_3(\mathcal{M}_q; \mathbb{Q}) \rightarrow H_3(X_s^b; \mathbb{Q})$ . With  $F \cdot \delta_q = 1$  as in the proof of Lemma 4.1, the restriction of  $\gamma$  is therefore  $(\gamma \cdot \delta_q)F$ , and  $\kappa_q$  is  $(\gamma \cdot \delta_q) \partial F$ . The same excision at a cone point  $p \notin \Sigma$ , followed by the isomorphism  $\partial: H_3(\mathcal{M}_p, L_p; \mathbb{Q}) \rightarrow \mathcal{R}_p$  of Lemma 4.1, gives the second statement.  $\square$

*Remark 4.5.* The construction of  $\Omega$  in Section 3.1 is unchanged in this setting. It uses the decomposition

$$X_s = W \cup \bigsqcup_{p \notin \Sigma} \mathcal{M}_p \cup \bigsqcup_{p \in \Sigma} (X_s \cap U_p)$$

along all the links, the vanishing  $H^1(L_p; \mathbb{Z}) = 0$  of Section 3.1, which covers the  $\mathbb{F}_1$  link, and the triviality of  $\omega_{\mathcal{X}/\mathbb{D}}$  near each  $(p, 0)$ . The fiber  $X_s$  is allowed to be singular. That it is Gorenstein needs an argument now that it is not smooth, and the following one replaces the appeal to base change:  $X_0$  is Gorenstein and is the divisor  $\{t = 0\}$  of  $\mathcal{X}$  with  $t$  a nonzerodivisor by flatness, so  $\mathcal{X}$  is Gorenstein near  $X_0$ ; each  $X_s$  is then a Cartier divisor in a Gorenstein space and hence Gorenstein, and adjunction makes  $\omega_{\mathcal{X}/\mathbb{D}}$  invertible with  $\omega_{\mathcal{X}/\mathbb{D}}|_{X_s} \cong \omega_{X_s}$ . The singularities of  $X_s$  are isolated for small  $s$ , since the singular locus of  $\mathcal{X} \rightarrow \mathbb{D}$  is closed, proper over  $\mathbb{D}$ , and meets  $X_0$  in a finite set; the same argument places  $\text{Sing } X_s$  inside  $\bigsqcup_p \text{int } U_p$ , which is what makes  $X_s^b$  lie in the smooth locus. Since  $X_s^b$  lies in the smooth locus of  $X_s$ , the period arguments of Lemmas 3.1 and 3.3 apply verbatim with  $H_3(X_s; \mathbb{Q})$  replaced by  $H_3(X_s^b; \mathbb{Q})$ : requiring a bounding 4-chain to lie in  $X_s^b$  is a stronger demand than requiring it in  $X_s$ , so the nonvanishing concluded is the weaker of the two and is what Section 7 uses; the cycles involved lie in  $X_s^b$  by construction. Poincaré–Lefschetz duality on the compact oriented manifold with boundary  $X_s^b$ , which pairs  $H_3(X_s^b; \mathbb{Q})$  perfectly with  $H_3(X_s^b, \partial X_s^b; \mathbb{Q})$ , replaces Poincaré duality on  $X_s$ .

## 5. THE DISTINGUISHED HYPERSURFACE AND ITS FINITE DATA

**5.1. The threefold.** Let  $\Delta = \Delta_{\mathbb{F}_1}$  be the 22-vertex reflexive 4-polytope of [Wue26c, Theorem 6.1] and  $\Delta^\vee$  its polar, with the 26 vertices  $q_1, \dots, q_{26}$  listed in Table 1. For a reflexive polytope  $\Pi$  we write  $\mathbb{P}_\Pi$  for the Gorenstein Fano toric fourfold whose fan is the face fan of  $\Pi$ ; its anticanonical polytope is the polar of  $\Pi$ , so  $\mathbb{P}_{\Delta^\vee}$  is the variety written  $\mathbb{P}_\Delta$  in Batyrev’s notation [Bat94]. A Batyrev-regular anticanonical hypersurface  $X^\circ \subset \mathbb{P}_{\Delta^\vee}$  has exactly 26 ordinary double points (one for each unit-square 2-face of  $\Delta^\vee$ ), two anticanonical dP<sub>7</sub>-cone points (the two pentagon faces with one interior point) and two dP<sub>6</sub>-cone points (the two hexagon faces). This is stated for a generic equation in [Wue26c, Section 6]; the following

argument applies to every Batyrev-regular equation. All thirty germs are *analytically* the cones: the ambient chart at the 3-dimensional cone over a 2-face  $F$  splits as  $C(F) \times \mathbb{C}^*$ , the restriction of the defining section to the one-dimensional orbit is  $t^k(c_a + c_b t)$  with  $c_a, c_b \neq 0$  by Batyrev regularity because the dual edge  $F^*$  has lattice length one, and Weierstrass preparation in the orbit coordinate exhibits the germ as a graph over  $(C(F), 0)$ . In particular each face contributes exactly one singular point, the simple zero of  $c_a + c_b t$ .

$X^\circ$  satisfies the standing hypotheses of Section 3: it is an anticanonical Cartier divisor in the Gorenstein Fano toric fourfold  $P = \mathbb{P}_{\Delta^\vee}$ , so  $\omega_{X^\circ} \cong \mathcal{O}_{X^\circ}$  by adjunction, and  $h^1(\mathcal{O}_{X^\circ}) = 0$  from the sequence  $0 \rightarrow \mathcal{O}_P(K_P) \rightarrow \mathcal{O}_P \rightarrow \mathcal{O}_{X^\circ} \rightarrow 0$  together with  $H^1(\mathcal{O}_P) = 0$  and  $H^2(\mathcal{O}_P(K_P)) \cong H^2(\mathcal{O}_P)^\vee = 0$  (toric vanishing and Serre duality on the complete toric fourfold).

**5.2. The resolution and the matrix.** Five terms recur in what follows, and we fix them here.

A *branch* at a singular point is a reduced irreducible component of the germ's miniversal base: a node has one, a dP<sub>7</sub> point one (the reduced line of Section 2.2), and a dP<sub>6</sub> point two, the  $A_1$  line and the  $A_2$  plane. A one-parameter smoothing of  $X$  induces at each point an arc into one branch, and  $\mathcal{R}_p$  depends on which. A *profile* for  $X$  is a choice of branch at every dP<sub>6</sub> point at once; the other germ types offer no choice, so  $X$  with  $k$  del Pezzo sextic points has  $2^k$  profiles, and a criterion that must hold for every smoothing is evaluated on each.

A coordinate of  $\bigoplus_p H_2(L_p; \mathbb{Q})$  is *covered* when Theorem 6.1 asserts nonvanishing there: every node coordinate, the  $(-2)$ -line at every dP<sub>7</sub> point, and the  $A_1$  summand at a dP<sub>6</sub> point on a profile that sends it to the line branch. The  $A_2$  plane is not covered, whose later treatment is described in Section 10.1.

The *root subspace* of a cone point is  $\mathcal{R}_p$  read in  $K^\perp$  through Proposition 2.1: the  $(-2)$ -line at a dP<sub>7</sub> point, and the  $A_1$  or the  $A_2$  summand of  $K^\perp \cong A_1 \oplus A_2$  at a dP<sub>6</sub> point. The name records that at degree 6 these are the root sublattices of  $K^\perp$ . At degree 7 the lattice  $K^\perp$  has discriminant 7 and is not generated by its roots, and the subspace is the line spanned by its unique  $(-2)$ -class up to sign.

We call the rows of  $B$  at a germ its *relation rows*; Proposition 8.2 shows that their complexifications form a basis of the germ's  $T^1$ .

Fix the crepant resolution  $\widehat{X} \rightarrow X^\circ$  obtained from the projective simplicial subdivision of the face fan of  $\Delta^\vee$  using all thirty boundary lattice points, with the star subdivision at the

TABLE 1. The 30 lattice points of  $\partial\Delta^\vee$ : the vertices  $q_1, \dots, q_{26}$  and the four face-interior points:  $q_{27}, q_{28}$  are the interior points of the two hexagons (below  $E_{6,1}, E_{6,2}$ ) and  $q_{29}, q_{30}$  of the two pentagons (below  $E_{7,1}, E_{7,2}$ ).

|                            |                           |                           |                          |
|----------------------------|---------------------------|---------------------------|--------------------------|
| $q_1 = (-1, -1, -1, -1)$   | $q_2 = (-1, -1, -1, 0)$   | $q_3 = (-1, -1, 0, -1)$   | $q_4 = (-1, -1, 0, 1)$   |
| $q_5 = (-1, -1, 1, 0)$     | $q_6 = (-1, -1, 1, 1)$    | $q_7 = (-1, 0, -1, -1)$   | $q_8 = (-1, 0, -1, 0)$   |
| $q_9 = (-1, 0, 0, -1)$     | $q_{10} = (-1, 0, 0, 1)$  | $q_{11} = (-1, 0, 1, 0)$  | $q_{12} = (-1, 0, 1, 1)$ |
| $q_{13} = (0, -1, -1, -1)$ | $q_{14} = (0, -1, -1, 0)$ | $q_{15} = (0, -1, 0, 1)$  | $q_{16} = (0, -1, 0, 0)$ |
| $q_{17} = (0, 1, -1, -1)$  | $q_{18} = (0, 1, -1, 0)$  | $q_{19} = (0, 1, 0, -1)$  | $q_{20} = (0, 1, 0, 0)$  |
| $q_{21} = (1, 1, -1, -1)$  | $q_{22} = (1, 0, -1, 0)$  | $q_{23} = (0, 0, 0, -1)$  | $q_{24} = (0, 0, 0, 1)$  |
| $q_{25} = (0, 0, 1, 0)$    | $q_{26} = (0, 0, 1, 1)$   | $q_{27} = (-1, -1, 0, 0)$ | $q_{28} = (-1, 0, 0, 0)$ |
| $q_{29} = (0, 0, -1, -1)$  | $q_{30} = (0, 0, -1, 0)$  |                           |                          |

four face-interior points and the twenty-six square diagonals

$$\begin{aligned} & (q_2, q_7), (q_2, q_{13}), (q_3, q_7), (q_4, q_8), (q_4, q_{14}), (q_5, q_9), (q_5, q_{13}), (q_5, q_{23}), (q_9, q_{23}), \\ & (q_6, q_{10}), (q_{10}, q_{15}), (q_6, q_{11}), (q_6, q_{16}), (q_6, q_{25}), (q_8, q_{17}), (q_9, q_{17}), (q_{12}, q_{18}), (q_{12}, q_{24}), \\ & (q_{12}, q_{19}), (q_{12}, q_{25}), (q_{14}, q_{16}), (q_{16}, q_{23}), (q_{16}, q_{26}), (q_{18}, q_{19}), (q_{20}, q_{24}), (q_{20}, q_{25}), \end{aligned}$$

labeled  $N_1, \dots, N_{26}$  in this order; the subdivision is smooth and projective. Over the nodes  $\pi$  is small with exceptional curves  $\Gamma_i$ , and over the cone points the exceptional divisors are toric surfaces  $E_{6,1}, E_{6,2} \cong \text{dP}_6$  and  $E_{7,1}, E_{7,2} \cong \text{dP}_7$ .

Write  $\widehat{\Sigma}$  for that subdivided fan and  $\mathbb{P}_{\widehat{\Sigma}} = \widehat{X}$ 's ambient fourfold  $\text{TV}(\widehat{\Sigma})$ ; the letter  $Y$  stays with the cone germ of Section 2. We record the two families of functionals on  $\text{Pic}(\mathbb{P}_{\widehat{\Sigma}})$  (equivalently, by restriction, on  $\text{Pic}(\widehat{X}) \otimes \mathbb{Q}$ , though only one containment is used below) whose kernel computation is the finite input. *Node rows*: for each square with cyclic vertices  $(a, b, c, d)$  and chosen diagonal  $(a, c)$ , the intersection functional of  $\Gamma_i$ , with values  $-1$  at  $D_a, D_c$ ,  $+1$  at  $D_b, D_d$  and  $0$  otherwise; it annihilates principal divisors because  $q_a + q_c = q_b + q_d$ . *Divisorial rows*: for each exceptional surface, the composition of the restriction  $\text{Pic}(\mathbb{P}_{\widehat{\Sigma}}) \rightarrow \text{Pic}(E)$  ( $D_j \mapsto$  the boundary curve of  $E$  for the rays of the face,  $D_{\text{int}} \mapsto K_E$ , all other rays  $\mapsto 0$ ) with a basis of the rank-2 or 3 lattice  $K^\perp \subset \text{Pic}(E)^\vee$ . In the bases of Table 2 the three  $\text{dP}_6$  functionals  $\varphi_{s_1}, \varphi_{s_2}, \varphi_{s_3}$  satisfy  $\varphi_{s_i}^2 = -2$ ,  $\varphi_{s_1} \cdot \varphi_{s_2} = 1$ ,  $\varphi_{s_3} \perp \varphi_{s_1}, \varphi_{s_2}$ , so  $\langle \varphi_{s_3} \rangle$  is the  $A_1$  summand and  $\langle \varphi_{s_1}, \varphi_{s_2} \rangle$  the  $A_2$  summand; the two  $\text{dP}_7$  functionals satisfy  $\varphi_\beta^2 = -2$ ,  $\varphi_\alpha^2 = -4$ , and  $\varphi_\beta$  spans the unique  $(-2)$ -line. Thus, if  $e_{p,i}$  is a chosen basis class of  $H_2(L_p; \mathbb{Q})$  and  $\iota_p: L_p \hookrightarrow \widehat{X}$  is the inclusion, our convention is

$$B_{(p,i),j} = D_j \cdot (\iota_p)_* e_{p,i}.$$

Rows are indexed by local homology classes and columns by ambient divisor classes; a relation with coefficient vector  $\kappa$  satisfies  $B^\top \kappa = 0$ . Stacking the 26 node rows and the ten divisorial rows gives an integral matrix  $B \in \text{Mat}_{36 \times 26}(\mathbb{Z})$  in the dual basis of the 26 divisor classes complementary to the unimodular cone  $(q_{15}, q_{22}, q_{24}, q_{26})$ .

**5.3. The ambient kernel computation.** Write coordinates on  $\mathbb{Q}^{36}$  as

$$(u_1, \dots, u_{26}; s_1, s_2, s_3 \text{ at } E_{6,1}; s_1, s_2, s_3 \text{ at } E_{6,2}; \alpha_1, \beta_1; \alpha_2, \beta_2).$$

TABLE 2. Exceptional-surface data. Bases of  $\text{Pic}(E)$ : boundary divisors at the rays listed; intersection matrices  $I$ ; canonical classes  $K$ ; and the  $K^\perp$  functionals in the dual bases. The two  $\text{dP}_6$  surfaces carry identical data at rays  $(q_3, q_4, q_5, q_6)$ , resp.  $(q_9, q_{10}, q_{11}, q_{12})$ .

| surface   | lattice data                                                                                                                                | $K^\perp$ rows                                                                                          |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| $E_{6,i}$ | $I_6 = \begin{pmatrix} -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & -1 \end{pmatrix}, \quad K = (-1, -1, -2, -2)$      | $\varphi_{s_1} = (0, 0, -1, 1)$<br>$\varphi_{s_2} = (-1, -1, 1, 0)$<br>$\varphi_{s_3} = (-1, 1, 1, -1)$ |
| $E_{7,1}$ | $I_{7,1} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \text{ at } (q_{13}, q_{17}, q_{21}), \quad K = (-1, -1, -2)$ | $\varphi_\alpha = (1, -1, 0)$<br>$\varphi_\beta = (1, 1, -1)$                                           |
| $E_{7,2}$ | $I_{7,2} = \begin{pmatrix} -1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \text{ at } (q_{14}, q_{18}, q_{22}), \quad K = (-1, -1, -2)$ | $\varphi_\alpha = (-2, 0, 1)$<br>$\varphi_\beta = (-1, -1, 1)$                                          |

The labels  $s_1, s_2, s_3$  are chosen to match those of Section 2.2: there  $\{s_1 = s_2 = 0\}$  cuts out the line component of the miniversal base and  $\{s_3 = 0\}$  the plane component, and here the same two equations cut out the  $A_1$  line and the  $A_2$  plane of  $K^\perp$ . These are different spaces and the coincidence of equations is a choice of labelling, not an identification. The following statements are exact integral computations from the data of Tables 1 and 2 and the subdivision of Section 5.2:

- (1) rank  $B = 21$  and  $\dim \ker(B^\top) = 15$ ; the node subsystem has rank 18, with coloops exactly  $N_9$  and  $N_{11}$ ; the unique row coloop of the full matrix is the  $\beta_1$ -row, i.e.  $\beta_1$  vanishes identically on  $\ker(B^\top)$ . This last statement is read in the basis of Table 2: replacing  $\varphi_\alpha$  by  $\varphi_\alpha + c\varphi_\beta$  shears the kernel coordinates and can destroy it. What is basis-free is the restricted statement of item (2), since once  $\alpha = 0$  the coordinate  $\beta$  does not depend on the choice of complement to the  $(-2)$ -line. The second proof of Theorem 6.5 uses item (1) in this fixed basis; item (2) gives the same conclusion in basis-free form.
- (2) On the *enlarged restricted kernel*  $\ker(B^\top) \cap \{\alpha_1 = \alpha_2 = 0\}$ , of dimension 13, the four coordinates  $u_9, u_{11}, \beta_1, \beta_2$  vanish identically. The four finer profiles obtained by additionally restricting each  $\text{dP}_6$  block to its  $A_1$  line ( $s_1 = s_2 = 0$ ) or its  $A_2$  plane ( $s_3 = 0$ ) have dimensions 9, 10, 10, 12 and force the same four coordinates, together with one  $s_3$  on each mixed profile. The last dimension is 12 and not 11 because the two  $s_3$ -coordinates are proportional on the enlarged restricted kernel: imposing either alone drops the dimension by one, and imposing the second drops it no further. That proportionality is also why a mixed profile, which deletes one  $s_3$ , forces the other.
- (3) The forcing is critical in the  $\text{dP}_7$  data: releasing either  $\alpha$ -coordinate destroys it.

By construction  $\ker(B^\top)$  contains the kernel of  $\bigoplus_p H_2(L_p; \mathbb{Q}) \rightarrow H_2(\hat{X}; \mathbb{Q})$  written in these coordinates: the node generators map to  $\pm[\Gamma_i]$  and the divisorial classes to their pushforwards from  $K^\perp \subset H_2(E_p)$ , and the entries of each row are the pairings of its local homology class with the ambient divisor classes. Only this containment is used: if the ambient Picard group under-detected  $H_2(\hat{X}; \mathbb{Q})$ , the ambient kernel would be larger than the topological one and every forced-zero statement would still bind. Equality in fact holds here ( $h^{1,1}(\hat{X}) = 26$ , the thirty boundary lattice points of  $\Delta^\vee$  less four), and holds for every polytope in the class considered in this paper, by Proposition 8.3.

## 6. PROOFS OF THE MAIN RESULTS

**Theorem 6.1** (mixed necessity). *Let  $X$  be a connected compact complex threefold with  $\omega_X \cong \mathcal{O}_X$  and  $h^1(\mathcal{O}_X) = 0$  whose singular points are ordinary double points and anticanonical cones over  $\text{dP}_6$  or  $\text{dP}_7$ , let  $\hat{X} \rightarrow X$  be a crepant resolution as in Section 1, and suppose  $X$  admits a one-parameter smoothing. With  $\mathcal{R}_p = \ker(H_2(L_p; \mathbb{Q}) \rightarrow H_2(\mathcal{M}_p; \mathbb{Q}))$  for the actual Milnor fibers of the induced local smoothings, set*

$$K(X) = \left\{ \kappa \in \ker \left( \bigoplus_p H_2(L_p; \mathbb{Q}) \rightarrow H_2(\hat{X}; \mathbb{Q}) \right) : \kappa_p \in \mathcal{R}_p \text{ at every cone point } p \right\}.$$

Then:

- (1) for every node  $q$  there is  $\kappa \in K(X)$  with  $\kappa_q \neq 0$ ;
- (2) for every  $\text{dP}_7$  point  $q$  there is  $\kappa \in K(X)$  whose  $q$ -component is a nonzero element of the  $(-2)$ -line  $\mathcal{R}_q$ ;

- (3) for every  $\mathrm{dP}_6$  point  $q$  smoothed through the one-parameter component of its miniversal base there is  $\kappa \in K(X)$  whose  $q$ -component is a nonzero element of  $\mathcal{R}_q$  (the  $A_1$  summand).

*Proof.* Fix small  $s \neq 0$ . (1) By Lemma 3.1,  $\delta_q \neq 0$  in  $H_3(X_s; \mathbb{Q})$ ; by Poincaré duality on the closed oriented  $X_s$  choose  $\gamma$  with  $\gamma \cdot \delta_q \neq 0$ ; by Lemma 4.1,  $\kappa = \partial\gamma$  lies in  $K(X)$  and its  $q$ -component is a nonzero multiple of  $\gamma \cdot \delta_q$ . (2),(3) By Lemma 3.3 the generator  $\nu$  of  $H_3(\mathcal{M}_q; \mathbb{Q})$  survives in  $H_3(X_s; \mathbb{Q})$ ; choose  $\gamma$  with  $\gamma \cdot \nu_* \neq 0$ ; by Lemma 4.1 the  $q$ -component of  $\partial\gamma$  is the image of a nonzero element of  $H_3(\mathcal{M}_q, L_q; \mathbb{Q})$  under the isomorphism onto  $\mathcal{R}_q$ , and  $\mathcal{R}_q$  is identified with the stated root subspace by Proposition 2.1, Section 2.4 and Lemma 4.2.  $\square$

*Remark 6.2.* Since a  $\mathbb{Q}$ -vector space is not the union of finitely many proper subspaces, the per-point conclusions of Theorem 6.1 combine: there is a *single* class  $\kappa \in K(X)$  simultaneously nonzero at every node, every  $\mathrm{dP}_7$  point, and every line-smoothed  $\mathrm{dP}_6$  point.

**Corollary 6.3.** *In the purely nodal case, Theorem 6.1(1) with Remark 6.2 states: any one-parameter smoothing forces a relation among the classes  $[\Gamma_i] \in H_2(\widehat{X}; \mathbb{Q})$  of the exceptional curves of a small resolution in which every node appears with nonzero coefficient.*  $\square$

**Proposition 6.4.** *If  $X^\circ$  admitted a one-parameter smoothing, the vanishing spheres of the nodes  $N_9$  and  $N_{11}$  would satisfy  $\delta_9 = \delta_{11} = 0$  in  $H_3(X_s; \mathbb{Q})$ .*

*Proof.* By Lemma 4.1 every boundary class  $\partial\gamma$  lies in  $K(X^\circ)$ ; its  $\mathrm{dP}_7$ -components lie on the  $(-2)$ -lines by Proposition 2.1, and Lemma 4.2 identifies these with the  $(-2)$ -lines of the resolution-side coordinates of Section 5.3, so  $\partial\gamma$  lies in the enlarged restricted kernel, on which  $u_9$  and  $u_{11}$  vanish identically (item (2) there). Since the  $u_q$ -component of  $\partial\gamma$  is a nonzero multiple of  $\gamma \cdot \delta_q$ , nondegeneracy of the intersection pairing gives  $\delta_9 = \delta_{11} = 0$ .  $\square$

**Theorem 6.5.**  *$X^\circ$  admits no one-parameter smoothing.*

*Proof.* Suppose one exists. Every germ of  $X^\circ$  is then smoothed; in particular the node  $N_9$  is, so Lemma 3.1 gives  $\delta_9 \neq 0$  in  $H_3(X_s; \mathbb{Q})$ , contradicting Proposition 6.4.

Alternatively, avoiding the nodes altogether: the point covered by  $E_{7,1}$  is smoothed through the unique component of its miniversal base, so Theorem 6.1(2) produces  $\kappa \in K(X^\circ)$  whose  $E_{7,1}$ -component is a nonzero element of the  $(-2)$ -line, i.e. (Lemma 4.2 and Table 2) has nonzero  $\beta_1$ -coordinate; but  $\beta_1$  vanishes identically already on the unrestricted  $\ker(B^T)$  (Section 5.3, item (1)).  $\square$

*Remark 6.6.* The two proofs are independent in the period lemma and in the kernel coordinate used (Lemma 3.1 with the vanishing of  $u_9$  in Section 5.3(2), versus Lemma 3.3 with the vanishing of  $\beta_1$  in Section 5.3(1), which item (2) restates in basis-free form), and share the membership and boundary-compatibility inputs (Lemmas 4.1 and 4.2) and the divisorial rows of Table 2.

**Corollary 6.7.** *Let  $X$  satisfy the standing hypotheses of Theorem 6.1 and admit no one-parameter smoothing. Then no fiber of a small representative of the miniversal deformation of  $X$  is smooth; consequently  $X$  is not smoothable over any base germ, and neither is any small deformation of  $X$ . Likewise, no formal arc in the miniversal deformation of  $X$  meets all the local smoothing loci. Here a formal arc  $\mathrm{Spf} \mathbb{C}[[t]] \rightarrow \mathrm{Def}(X)$  meets the smoothing locus of a singular point  $p$  if the zeroth Fitting ideal of the coordinate ring of the relative critical locus of the induced deformation of  $(X, p)$  over  $\mathbb{C}[[t]]$ , a finite  $\mathbb{C}[[t]]$ -module, is nonzero, that is, if that deformation has smooth generic fiber. In particular all of this holds for  $X^\circ$ , by Theorem 6.5.*

*Proof.* The miniversal deformation of the compact  $X$  exists [Gra74] and its discriminant is a closed analytic subset of the base. If some fiber arbitrarily close to the origin were smooth, the curve selection lemma would produce a real-analytic arc through the origin with smooth generic fiber (the complement of the discriminant is semianalytic, so by [Loj65, §19, Prop. 2] it contains a semianalytic arc ending at the origin, which by [BM88, Lemma 6.3(1)] is the image of  $(0, \varepsilon)$  under a real-analytic map on  $(-\varepsilon, \varepsilon)$ ; for real algebraic sets see [Mil68]); complexifying (a real-analytic function vanishing on a real segment of a disk vanishes identically, and the discriminant preimage of the complexified arc is discrete) gives a holomorphic disk, that is, a one-parameter smoothing, contrary to hypothesis. Small deformations are covered by openness of versality. For the formal statement, suppose a formal arc meets every local smoothing locus, and let  $k$  be the largest  $t$ -adic order of the zeroth Fitting ideals in the definition. By Artin approximation [Art68] applied to the analytic miniversal deformation, some analytic arc agrees with the formal one modulo  $t^{k+1}$ . Fitting ideals commute with base change, so the corresponding Fitting ideals of the analytic arc agree with those of the formal arc modulo  $t^{k+1}$  and are therefore nonzero: for small  $t \neq 0$  the fibers of the analytic arc are smooth near every singular point of  $X$ . Since the singular locus of the fibers is closed in the total space and meets the central fiber only in  $\text{Sing } X$ , these fibers are, for small  $t$ , singular at most inside the Milnor-ball neighborhoods of those points, hence smooth everywhere, so the arc is a one-parameter smoothing, again contrary to hypothesis.  $\square$

*Remark 6.8.* The two nodal coloops quantify how far  $X^\circ$  is from the nodal theory: no relation among the twenty-six exceptional curves alone is nonzero at  $N_9$  or  $N_{11}$ , so a smoothing would have had to route through the divisorial directions, and items (2)–(3) of Section 5.3 say the divisorial directions cannot supply what is needed at any order. The mirror partner of  $X^\circ$  is treated in Section 7.

## 7. THE MIRROR PARTNER

The polytope  $\Delta$  itself carries a Calabi–Yau hypersurface, the mirror partner of  $X^\circ$  in the doubly isolated pair. Write  $X_\Delta \subset \mathbb{P}_\Delta$  for a Batyrev-regular anticanonical hypersurface, with the convention of Section 5.1 ( $\mathbb{P}_\Delta$  has the face fan of  $\Delta$  as its fan and  $\Delta^\vee$  as its anticanonical polytope), and  $v_1, \dots, v_{22}$  for the vertices of  $\Delta$  in the order of [Wue26c, Theorem 6.1], reproduced in Table 3. By that theorem  $X_\Delta$  has twenty ordinary double points, one  $\text{dP}_6$ -cone point over the hexagonal face  $\text{conv}\{v_9, v_{14}, v_{15}, v_{20}, v_{21}, v_{22}\}$ , and one point whose germ is the anticanonical cone over  $\mathbb{F}_1$ , over the quadrilateral face  $\text{conv}\{v_4, v_{18}, v_{19}, v_{22}\}$ . As in Section 5.1,  $X_\Delta$  is an anticanonical Cartier divisor in a Gorenstein Fano toric fourfold, so  $\omega_{X_\Delta} \cong \mathcal{O}_{X_\Delta}$  and  $h^1(\mathcal{O}_{X_\Delta}) = 0$ , and every germ is analytically the cone over its face because every dual

TABLE 3. The 24 boundary lattice points of  $\Delta = \Delta_{\mathbb{F}_1}$ : the vertices  $v_1, \dots, v_{22}$  in the order of [Wue26c, Theorem 6.1], the interior point  $v_{23}$  of the hexagonal face and the interior point  $v_{24}$  of the  $\mathbb{F}_1$  face.

|                            |                            |                           |                           |
|----------------------------|----------------------------|---------------------------|---------------------------|
| $v_1 = (1, 0, 0, 0)$       | $v_2 = (0, 1, 0, 0)$       | $v_3 = (1, -1, 0, 0)$     | $v_4 = (0, 0, 1, 0)$      |
| $v_5 = (0, 0, 0, 1)$       | $v_6 = (0, 0, 1, -1)$      | $v_7 = (0, 0, -1, 1)$     | $v_8 = (0, 0, 0, -1)$     |
| $v_9 = (0, 0, -1, 0)$      | $v_{10} = (-1, 1, 0, 0)$   | $v_{11} = (0, -1, 0, 0)$  | $v_{12} = (-1, 1, -1, 1)$ |
| $v_{13} = (0, -1, 0, -1)$  | $v_{14} = (0, -1, -1, 0)$  | $v_{15} = (-1, 1, -1, 0)$ | $v_{16} = (-1, 0, 0, -1)$ |
| $v_{17} = (-1, 0, -1, 1)$  | $v_{18} = (-1, -1, 0, -1)$ | $v_{19} = (-2, 1, -1, 1)$ | $v_{20} = (-2, 1, -1, 0)$ |
| $v_{21} = (-1, -1, -1, 0)$ | $v_{22} = (-2, 0, -1, 0)$  | $v_{23} = (-1, 0, -1, 0)$ | $v_{24} = (-1, 0, 0, 0)$  |

TABLE 4. The two exceptional surfaces of  $\widehat{X}_\Delta$ : the cyclic boundary rays, the intersection matrix  $I$  in the Picard basis of the boundary divisors at the rays listed, the canonical class  $K$ , and the  $K^\perp$  functionals in the dual basis. As in Table 2,  $\varphi_{s_3}$  spans the  $A_1$  summand and  $\varphi_{s_1}, \varphi_{s_2}$  the  $A_2$  summand, here with  $\varphi_{s_1} \cdot \varphi_{s_2} = -1$ . The  $\mathbb{F}_1$  lattice  $K^\perp$  has no  $(-2)$ -class.

| surface                                                                                  | lattice data                                                                                                                                                              | $K^\perp$ rows                                                                                         |
|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| $E_{6,\Delta}$ over $v_{23}$ , rays<br>( $v_{21}, v_{14}, v_9, v_{15}, v_{20}, v_{22}$ ) | $I_6 = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 1 & -1 & 1 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix}$ at $(v_9, v_{15}, v_{20}, v_{22})$ ,<br>$K = (-1, -2, -2, -1)$ | $\varphi_{s_1} = (1, -1, 0, 1)$<br>$\varphi_{s_2} = (0, -1, 1, 0)$<br>$\varphi_{s_3} = (1, -1, 1, -1)$ |
| $E_{\mathbb{F}_1}$ over $v_{24}$ , rays<br>( $v_{18}, v_4, v_{19}, v_{22}$ )             | $I_{\mathbb{F}_1} = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}$ at $(v_{19}, v_{22})$ ,<br>$K = (-3, -2)$                                                              | $\varphi_t = (-2, 3)$                                                                                  |

edge has lattice length one. The last germ is reduced-rigid, so  $X_\Delta$  admits no smoothing [Wue26a]. The reduced-rigid  $\mathbb{F}_1$  germ already prevents a smoothing of  $X_\Delta$ . Independently, Theorem 7.1 shows that the global obstruction is also present on  $X_\Delta$  and does not use that germ.

**Theorem 7.1** (necessity with germs left singular). *Let  $X$  be a connected compact complex threefold with  $\omega_X \cong \mathcal{O}_X$  and  $h^1(\mathcal{O}_X) = 0$  whose singular points are ordinary double points, anticanonical cones over  $\mathrm{dP}_6$  or  $\mathrm{dP}_7$ , and anticanonical cones over  $\mathbb{F}_1$ ; let  $\widehat{X} \rightarrow X$  be a crepant resolution as in Section 1. Let  $\mathcal{X} \rightarrow \mathbb{D}$  be any one-parameter deformation of  $X$  and let  $\Sigma$  be the set of points at which the nearby fiber is singular, a set containing every  $\mathbb{F}_1$ -cone point by Lemma 4.3. Put*

$$K^\flat(X; \Sigma) = \left\{ \kappa \in \ker \left( \bigoplus_p H_2(L_p; \mathbb{Q}) \rightarrow H_2(\widehat{X}; \mathbb{Q}) \right) : \kappa_p \in \mathcal{R}_p \text{ at every cone point } p \notin \Sigma \right\}.$$

*Then  $K^\flat(X; \Sigma)$  contains a class nonzero at  $q$  for every node  $q \notin \Sigma$ , for every  $\mathrm{dP}_7$  point  $q \notin \Sigma$ , and for every  $\mathrm{dP}_6$  point  $q \notin \Sigma$  smoothed through the one-parameter component of its miniversal base. For  $\Sigma = \emptyset$  this is Theorem 6.1.*

*Proof.* The proof of Theorem 6.1 applies with Remark 4.5 and Lemma 4.4 in place of Lemma 4.1. At a node  $q \notin \Sigma$ , Lemma 3.1 gives  $\delta_q \neq 0$  in  $H_3(X_s^b; \mathbb{Q})$ , since a bounding 4-chain in  $X_s^b$  would make the period of the closed form  $\Omega_s$  vanish; Poincaré–Lefschetz duality on the compact oriented manifold with boundary  $X_s^b$  produces  $\gamma \in H_3(X_s^b, \partial X_s^b; \mathbb{Q})$  with  $\gamma \cdot \delta_q \neq 0$ ; and Lemma 4.4 converts  $\gamma$  into a class  $\kappa \in K^\flat(X; \Sigma)$  with  $\kappa_q \neq 0$ . At the cone points  $q \notin \Sigma$  in the two covered branches the same substitution is made in the argument of Theorem 6.1(2),(3), with Lemma 3.3 read in  $H_3(X_s^b; \mathbb{Q})$ .  $\square$

Whatever the deformation and whatever  $\Sigma$ , the space  $K^\flat(X; \Sigma)$  is by definition contained in  $\ker \left( \bigoplus_p H_2(L_p; \mathbb{Q}) \rightarrow H_2(\widehat{X}; \mathbb{Q}) \right)$ , so a coordinate that vanishes identically on the latter is forced in every case. On  $X_\Delta$  that is what happens.

Fix the smooth projective crepant subdivision of the face fan of  $\Delta$  that uses all twenty-four boundary lattice points, with the star subdivision at  $v_{23}$  and  $v_{24}$  and the twenty square

diagonals

$$\begin{aligned} &(v_2, v_7), (v_2, v_9), (v_3, v_8), (v_3, v_9), (v_{10}, v_{12}), (v_{10}, v_{15}), (v_{11}, v_{13}), (v_{11}, v_{14}), (v_{11}, v_{17}), \\ &(v_{10}, v_{16}), (v_9, v_{12}), (v_{12}, v_{17}), (v_{14}, v_{17}), (v_9, v_{13}), (v_{13}, v_{16}), (v_{15}, v_{16}), (v_{15}, v_{19}), (v_{14}, v_{18}), \\ &(v_{18}, v_{20}), (v_{19}, v_{21}), \end{aligned}$$

labeled  $n_1, \dots, n_{20}$ ; the subdivision is smooth, with 94 maximal cones, and projective. The corresponding circuit relations are

$$\begin{aligned} v_1 + v_{12} &= v_2 + v_7, & v_1 + v_{15} &= v_2 + v_9, & v_1 + v_{13} &= v_3 + v_8, & v_1 + v_{14} &= v_3 + v_9, \\ v_2 + v_{19} &= v_{10} + v_{12}, & v_2 + v_{20} &= v_{10} + v_{15}, & v_3 + v_{18} &= v_{11} + v_{13}, & v_3 + v_{21} &= v_{11} + v_{14}, \\ v_5 + v_{21} &= v_{11} + v_{17}, & v_6 + v_{20} &= v_{10} + v_{16}, & v_7 + v_{15} &= v_9 + v_{12}, & v_7 + v_{19} &= v_{12} + v_{17}, \\ v_7 + v_{21} &= v_{14} + v_{17}, & v_8 + v_{14} &= v_9 + v_{13}, & v_8 + v_{18} &= v_{13} + v_{16}, & v_8 + v_{20} &= v_{15} + v_{16}, \\ v_{12} + v_{20} &= v_{15} + v_{19}, & v_{13} + v_{21} &= v_{14} + v_{18}, & v_{16} + v_{22} &= v_{18} + v_{20}, & v_{17} + v_{22} &= v_{19} + v_{21}, \end{aligned}$$

in the order  $n_1, \dots, n_{20}$ ; in each of them the right-hand pair is the chosen diagonal, so the circuit row of Section 5.2 is  $-1$  there and  $+1$  on the left-hand pair. Choosing the other diagonal negates the row and changes nothing below. Table 4 records the two exceptional surfaces. The lattice  $K^\perp(\mathbb{F}_1)$  is a line spanned by a primitive class of square  $-8$ , and it contains no  $(-2)$ -class, consistently with the reduced-rigidity of that germ. No condition is imposed at that point in any case: by Lemma 4.3 it lies in  $\Sigma$  for every deformation, so no Milnor fiber and no  $\mathcal{R}_p$  is defined there. Stacking the twenty node rows, the three hexagon rows and the single  $\mathbb{F}_1$  row gives an integral matrix  $B_\Delta \in \text{Mat}_{24 \times 24}(\mathbb{Z})$ , the columns being indexed by the divisors of the twenty-four boundary lattice points, which span  $\text{Pic}$  of the resolution without being a basis. The following are exact integral computations, reproduced by the ancillary programs from the printed data of Tables 3 and 4:

- (1)  $\text{rank } B_\Delta = 16$  and  $\dim \ker(B_\Delta^\top) = 8$ ; the node subsystem has rank 14, with coloops exactly  $n_9$  and  $n_{10}$ ;
- (2) on the unrestricted kernel  $\ker(B_\Delta^\top)$  the coordinates of  $n_9$ , of  $n_{10}$ , of the  $\mathbb{F}_1$  row and of the  $A_1$  row of the hexagon vanish identically, so no branch restriction is needed;
- (3) the column of  $B_\Delta$  at the divisor of  $v_{24}$  is identically zero, the node rows having no entry there and the divisorial rows having entry  $\langle \varphi, K_E \rangle = 0$ ; the same holds at  $v_{23}$ . Discarding the  $\mathbb{F}_1$  row and that column altogether leaves a system on which  $n_9$  and  $n_{10}$  are still forced.

**Theorem 7.2.** *No one-parameter deformation of  $X_\Delta$  has nearby fiber smooth at the node  $n_9 = \text{conv}\{v_5, v_{11}, v_{17}, v_{21}\}$ , and none has nearby fiber smooth at  $n_{10} = \text{conv}\{v_6, v_{10}, v_{16}, v_{20}\}$ . The same two conclusions hold for deformations over arbitrary analytic base germs. In particular no analytic deformation of  $X_\Delta$  smooths all twenty nodes and the  $\text{dP}_6$  point, whatever it does at the  $\mathbb{F}_1$  point. By item (2) and Theorem 7.1, the same argument applied to the  $A_1$  coordinate shows that no analytic deformation of  $X_\Delta$  smooths its  $\text{dP}_6$  point through the one-parameter component of its miniversal base. The reduced-rigidity of the  $\mathbb{F}_1$  germ excludes the smoothings of  $X_\Delta$ ; the deformations excluded here are ones that it leaves available.*

*Proof.* Suppose the nearby fiber is smooth at  $n_9$ , so  $n_9 \notin \Sigma$ . By Theorem 7.1 there is  $\kappa \in K^\flat(X_\Delta; \Sigma)$  with  $\kappa_{n_9} \neq 0$ . But  $K^\flat(X_\Delta; \Sigma)$  is contained in  $\ker(B_\Delta^\top)$  for every  $\Sigma$ , read in the coordinates above, and the  $n_9$  coordinate vanishes identically there. The argument for  $n_{10}$  is the same.

For the  $\mathrm{dP}_6$  assertion, suppose first that the base is a disk and that the local classifying arc lies in the one-parameter component of the miniversal base. Theorem 7.1 then gives a class whose  $A_1$  coordinate is nonzero, contrary to item (2).

Now let the deformation have an arbitrary analytic base germ  $(S, 0)$ . Since the assertions concern actual nearby fibers, we may replace  $S$  by  $S_{\mathrm{red}}$ . If one of the excluded local smoothings occurred arbitrarily close to the origin, the curve-selection argument used in Corollary 6.7, applied to the complement of the inverse image of the relevant local discriminant, would give a holomorphic disk through the origin whose punctured fibers have the same local smoothing property. For the  $\mathrm{dP}_6$  assertion, apply curve selection inside the inverse image of the one-parameter component and outside its discriminant. Pullback to the resulting disk contradicts the one-parameter conclusions just proved.  $\square$

**Corollary 7.3.** *Both members of the doubly isolated mirror pair  $(X^\circ, X_\Delta)$  carry the obstruction of this paper: on  $X^\circ$ , in the form of Theorem 6.1, it excludes every smoothing; on  $X_\Delta$ , in the form of Theorem 7.1, it excludes every analytic deformation smoothing  $n_9$  or  $n_{10}$ . On  $X_\Delta$  the two mechanisms are independent: by item (2) the forcing holds on the unrestricted kernel, where no local condition is imposed at the  $\mathbb{F}_1$  point, and by item (3) it persists when the  $\mathbb{F}_1$  data are discarded altogether, whereas the reduced-rigidity of that germ is a statement about it alone.*

## 8. WHAT THE CRITERION MEASURES

The rows of Section 5.2 were introduced as a device: the  $K^\perp$  functionals stand in for the exceptional curve class that a del Pezzo cone point does not have. This section identifies their span with a rational form of the germ's  $T^1$ . It also makes the finite test of Section 9 readable and locates the criterion against the smoothing theorems of Gross [Gro97].

**8.1. The combinatorial setting.** Throughout this section and the next,  $\Pi \subset N_{\mathbb{R}} \cong \mathbb{R}^4$  is a reflexive 4-polytope with polar  $\Pi^*$ ,  $\mathbb{P}_\Pi$  is the Gorenstein Fano toric fourfold of its face fan, and  $X \subset \mathbb{P}_\Pi$  is an anticanonical hypersurface regular in Batyrev's sense [Bat94, Definition 3.1.1]. (The letter  $\Delta$  stays reserved for the fixed  $\Delta_{\mathbb{F}_1}$  of Section 5.1; the polytope relevant to  $X^\circ$  is  $\Pi = \Delta^\vee$ , and the polytope relevant to  $X_\Delta$  is  $\Pi = \Delta$ .)

For a 2-face  $G$  of  $\Pi$  with dual edge  $G^* \subset \Pi^*$  of lattice length  $\ell(G^*)$ , the hypersurface meets the closed orbit of  $\mathrm{Cone}(G)$  in  $\ell(G^*)$  points, and at each of them the germ of  $X$  is the affine toric threefold of  $\mathrm{Cone}(G)$ , exactly as in Section 5.1 [Bat94], [Wue26a, Proposition 3.1]. The germ is singular unless  $G$  is a unimodular triangle, and it is isolated when every edge of  $G$  has lattice length one; we call such a  $G$  a *singular 2-face* and write  $(Y_G, 0)$  for the germ. A unit square gives a node; a pentagon with unit edges and one interior lattice point gives  $C(\mathrm{dP}_7)$  and a hexagon with unit edges and one interior lattice point gives  $C(\mathrm{dP}_6)$ , the interior point being the origin of the fan of the exceptional surface in each case. (A polygon with exactly one interior lattice point is reflexive, and among the sixteen reflexive polygons the only ones with five or six vertices and no further boundary lattice points are the fan polygons of  $\mathrm{dP}_7$  and  $\mathrm{dP}_6$ .)

**Definition 8.1.**  $\Pi$  is *admissible* if every edge of  $\Pi$  has lattice length one; every 2-face of  $\Pi$  has either three vertices and no interior lattice point, or four vertices and no interior lattice point, or five or six vertices and exactly one interior lattice point; and  $\ell(G^*) = 1$  for every singular 2-face  $G$ .

The four types are the unimodular triangle, the unit square, the pentagon and the hexagon listed above: with unit edges, a quadrilateral without interior points has normalized area 2

by Pick's formula, so both triangles cut off by a diagonal are unimodular, and convexity then forces  $v_a + v_c = v_b + v_d$ : it is a unimodular parallelogram. Note that a quadrilateral with unit edges and *one* interior point is a different face, giving a cone point over a smooth toric del Pezzo surface of degree 8, either  $\mathbb{F}_1$  or  $\mathbb{P}^1 \times \mathbb{P}^1$ ; both are excluded here, and the  $\mathbb{F}_1$  case is the one that occurs on  $\Delta$  and is treated in Section 7.

For admissible  $\Pi$  the hypersurface  $X$  satisfies the standing hypotheses of Theorem 6.1, by the argument of Section 5.1:  $\text{Sing } X$  is finite with one point per singular 2-face, and every germ is a node or an anticanonical  $\text{dP}_6$  or  $\text{dP}_7$  cone. The crepant resolution is taken throughout from a maximal projective crepant partial desingularization of  $\mathbb{P}_\Pi$  [Bat94, §2.2] whose fan refines the star subdivision of the cone over each pentagonal or hexagonal 2-face at its interior lattice point, restricted to  $X$  [Bat94, §4.2]. Such a desingularization exists. Pulling the origin and then the interior points of these faces gives a regular subdivision of the lattice points of  $\Pi$  [DLRS10, Lemma 4.3.12] whose restriction to each such face is the star subdivision; it need not use the lattice points interior to facets. Place every lattice point on the lower envelope of a height function inducing this subdivision and perturb by  $\varepsilon$  times a strictly convex function and then generically: for small  $\varepsilon$  this gives a regular triangulation that refines the subdivision and uses every lattice point [DLRS10, Lemma 2.3.16]. Every maximal simplex contains the origin, the only interior lattice point of  $\Pi$ , and the triangles  $(c, v_j, v_{j+1})$ , with  $c$  the interior point of the face, are unimodular and are not subdivided, so the triangulation still restricts to the star subdivision on each such face. The resolution is then small over the nodes and is the blowup of the vertex at each cone point, as Theorem 6.1 requires. An arbitrary maximal triangulation need not have this property: it may cut off a unimodular corner of a pentagon or hexagon (Section 4.2). The polytope  $\Delta^\vee$  of Section 5.1 is admissible, its 68 two-faces being 38 unimodular triangles, 26 unit squares, two pentagons and two hexagons;  $\Delta_{19}$  and  $\Delta_{20}$  of Section 9 are admissible;  $\Delta$  itself is not, because the quadrilateral face carrying its  $\mathbb{F}_1$ -cone point is neither a square nor a triangle, but it satisfies the weaker hypothesis of Proposition 8.3 below.

## 8.2. The relation rows are the germ's $T^1$ .

**Proposition 8.2.** *Let  $G$  be a singular 2-face of  $\Pi$  with unit edges and vertices  $v_1, \dots, v_n$  in cyclic order, and put*

$$R_G := \left\{ a \in \mathbb{Q}^n : \sum_j a_j v_j = 0 \right\}.$$

*Then  $R_G$  has dimension  $n - 3$ , Altmann's model gives a natural isomorphism*

$$T_{(Y_G, 0)}^1 \cong R_G \otimes_{\mathbb{Q}} \mathbb{C}.$$

*If  $G$  is a unit square, or a pentagon or hexagon with one interior lattice point, the rows of Section 5.2 at  $G$ , read in the coordinates of its rays, form a  $\mathbb{Q}$ -basis of  $R_G$ . In the pentagon and hexagon cases, the crepant resolution has an exceptional surface  $E$  over the germ, and  $R_G$  is  $K_{\mathbb{Q}}^\perp \subset \text{Pic}(E)_{\mathbb{Q}}^\vee$ ; when  $G$  is a unit square it is the line spanned by the intersection vector of the exceptional curve of a small resolution.*

*Proof.* Altmann shows that  $T^1$  of the germ is concentrated in the Gorenstein degree [Alt95, (6.5) and the subsequent Remark] and identifies it with the complex vector space of dilations of the edges of  $G$  satisfying the closing condition, modulo the homothety [Alt97, (2.2), (2.6)]; its dimension is  $(n - 2) - 1$ . Send a dilation  $t$ , with  $t_j$  the dilation of the edge  $(v_j, v_{j+1})$ , to its jump  $a_j := t_j - t_{j-1}$  across the vertex  $v_j$ . Then

$$\sum_j a_j v_j = \sum_j t_j v_j - \sum_j t_j v_{j+1} = - \sum_j t_j (v_{j+1} - v_j) = 0$$

by the closing condition; the kernel of  $t \mapsto a$  is the constants. The construction is defined over  $\mathbb{Q}$ , and its rational form maps isomorphically to  $R_G$ : both spaces have dimension  $n - 3$  because  $\text{Cone}(G)$  is three-dimensional. Extending scalars gives the stated isomorphism with  $T^1$ . For the last two clauses, note first that  $G$  lies in the affine hyperplane  $\langle u_F, \cdot \rangle = -1$  for either facet  $F \supset G$ , so  $\sum_j a_j v_j = 0$  forces  $\sum_j a_j = 0$ . If  $G$  is a pentagon or hexagon, the  $v_j$  are the rays of the fan of  $E$  seen from the interior lattice point, and dualizing  $0 \rightarrow M_E \rightarrow \mathbb{Q}^n \rightarrow \text{Pic}(E)_{\mathbb{Q}} \rightarrow 0$  identifies  $\{a : \sum_j a_j v_j = 0 \text{ in the plane of } G\}$  with  $\text{Pic}(E)_{\mathbb{Q}}^{\vee}$ ; since  $K_E = -\sum_j D_j$  the extra condition  $\sum_j a_j = 0$  is orthogonality to  $K_E$ , and a divisorial row of Section 5.2 is  $\varphi \mapsto (\langle \varphi, D_j \rangle)_j$  for  $\varphi$  in a basis of  $K^{\perp}$ , which is that identification. If  $G$  is a unit square with cyclic vertices  $(a, b, c, d)$ , the space is one-dimensional and spanned by  $(-1, 1, -1, 1)$ , because  $v_a + v_c = v_b + v_d$ ; that is the node row of Section 5.2, the intersection functional of  $\Gamma$ .  $\square$

Thus the  $K^{\perp}$  lattice introduced in Section 5.2 as a substitute for the missing exceptional curve class is a natural rational form of the tangent space to the germ's deformation functor. A linear relation among the rays of a cone is the combinatorial avatar of a curve class, with  $a_j$  the intersection number against the invariant divisor  $D_j$ ; at a node the square's relation  $v_a + v_c = v_b + v_d$  is the class of the exceptional curve of a small resolution, and the row  $(-1, 1, -1, 1)$  of Section 5.2 is its vector of intersection numbers. The proposition says that this reading survives at a del Pezzo cone point, where the curve is not there to be read. Two consequences follow. The interior lattice point of a pentagon or hexagon carries coefficient  $\langle \varphi, K \rangle = 0$  in every row, so the rows are supported on the vertices of the face; and the total number of relation rows equals  $\dim H^0(\mathcal{T}_X^1)$ , which is 36 for  $X^{\circ}$  and 16 for  $X_{19}$ .

### 8.3. The ambient kernel is the topological kernel.

**Proposition 8.3.** *Suppose every 2-face of  $\Pi$  is either a unimodular triangle or has dual edge of lattice length one. Let  $\hat{X} \rightarrow X$  be a smooth hypersurface resolution induced by a maximal projective crepant partial desingularization of  $\mathbb{P}_{\Pi}$ . Write  $\ell^*(G)$  for the number of lattice points in the relative interior of  $G$ , and  $\ell(G^*)$  for the lattice length of the dual edge. Then the correction term  $\sum_G \ell^*(G)(\ell(G^*) - 1)$ , summed over 2-faces  $G$  of  $\Pi$ , in Batyrev's formula [Bat94, Corollary 4.5.1] for  $h^{1,1}$  vanishes, and the ambient divisor classes span  $H^2(\hat{X}; \mathbb{Q}) = H^{1,1}(\hat{X}; \mathbb{Q})$ .*

*Suppose in addition that  $X$  has only isolated singularities, all nodes or anticanonical cones over  $\text{dP}_6$ ,  $\text{dP}_7$  or  $\mathbb{F}_1$ , and that  $\hat{X} \rightarrow X$  is small over the nodes and is the blowup of the vertex at each cone point. Define  $B$  by the pairings of bases of  $H_2(L_p; \mathbb{Q})$  with ambient divisor classes, as in Sections 5.2 and 7. Then*

$$\ker(B^{\top}) = \ker\left(\bigoplus_p H_2(L_p; \mathbb{Q}) \longrightarrow H_2(\hat{X}; \mathbb{Q})\right).$$

*These additional hypotheses hold for admissible polytopes with the resolution of Section 8.1 and for  $\Delta$  with the resolution of Section 7.*

*Proof.* A unimodular triangle has  $\ell^*(G) = 0$ ; every other 2-face has  $\ell(G^*) - 1 = 0$  by hypothesis. Hence every summand of the correction term vanishes. In the proof of [Bat94, Theorem 4.4.2], the divisor  $D_{\rho}$  indexed by an interior lattice point of  $G$  meets  $\hat{X}$  in  $\ell(G^*)$  irreducible components. Thus  $\ell^*(G)(\ell(G^*) - 1)$  counts the additional components beyond one for each such ambient divisor. When the correction vanishes, every nonempty restriction of a torus-invariant divisor is irreducible. These restrictions span  $H^2(\hat{X}; \mathbb{Q})$ . Indeed, let

$Z \subset \hat{X}$  be the part lying over the open torus  $T \cong (\mathbb{C}^*)^4$ , a nondegenerate affine hypersurface, and let  $D = \hat{X} \setminus Z$ , a divisor with simple normal crossings whose components are those restrictions. By the Lefschetz theorem of Danilov–Khovanskii [DK86, Proposition 3.9 and Remark 3.10] the restriction  $H^2(T; \mathbb{Q}) \rightarrow H^2(Z; \mathbb{Q})$  is an isomorphism, so  $H^2(Z; \mathbb{Q})$  is pure of weight 4. The image of the pure weight-2 space  $H^2(\hat{X}; \mathbb{Q})$  in it is therefore zero, so by the exact sequence  $H_D^2(\hat{X}; \mathbb{Q}) \rightarrow H^2(\hat{X}; \mathbb{Q}) \rightarrow H^2(Z; \mathbb{Q})$ , in which  $H_D^2(\hat{X}; \mathbb{Q}) \cong H_4(D; \mathbb{Q})$  is spanned by the components of  $D$ , the space  $H^2(\hat{X}; \mathbb{Q})$  is spanned by their classes. The smooth Calabi–Yau hypersurface  $\hat{X}$  has  $h^{2,0}(\hat{X}) = 0$ , so  $H^2(\hat{X}; \mathbb{Q}) = H^{1,1}(\hat{X}; \mathbb{Q})$ .

Under the additional isolated-germ hypotheses, a row of  $B$  records the pairings of a class with the ambient divisors, so a combination of rows vanishes exactly when the corresponding total class pairs to zero against every ambient divisor. The evaluation pairing  $H_2 \times H^2 \rightarrow \mathbb{Q}$  is perfect; the ambient divisor classes span  $H^2$  by the first part, and therefore the total class pairs to zero against all of them exactly when it vanishes in  $H_2(\hat{X}; \mathbb{Q})$ .  $\square$

The correction-term hypothesis alone does not imply that  $X$  has isolated singularities; the additional assumptions are needed to interpret the matrix as the isolated-link pairing used in this paper. Section 5.3 uses only the containment: a coordinate vanishing on the ambient kernel also vanishes on the topological kernel. The proposition gives equality for the admissible class and the treated mirror partner. This equality is needed when applying the converse criterion of the companion paper [Wue26e] to the matrix kernel. The correction-term vanishing is distinct from the one that empties Mavlyutov’s mechanism of non-polynomial deformations [Mav04]: that is the mirror correction, summed over 2-faces of  $\Pi^*$ , and it follows from the unit edges of  $\Pi$ . Both vanishings hold for the admissible polytopes and for  $\Delta$ .

**8.4.  $\mathbb{Q}$ -factoriality and primitive contractions.** Gross proves that a primitive contraction of a divisor to a point on a smooth Calabi–Yau threefold has smoothable target unless the exceptional divisor is  $\mathbb{P}^2$  or  $\mathbb{F}_1$  [Gro97, Theorem 5.8]; degrees 6 and 7 are allowed. Namikawa proves that a  $\mathbb{Q}$ -factorial Calabi–Yau threefold with isolated toric Gorenstein singularities, possibly several, has surjective map  $\text{Def}(X) \rightarrow \prod_p \text{Def}(X, p)$ , and hence is smoothable as soon as its germs are [Nam97, Corollary 6]. When a smooth Calabi–Yau threefold contains a del Pezzo surface  $E$  of degree 6 whose contraction alone produces a single singular point  $p$ , he shows that the image of the restriction of divisors to  $E$  determines the image of the deformations of the contracted threefold in the deformation space of its germ at  $p$ , and in particular which smoothing components of the germ that image contains [Nam97, Theorem 10]; the divisorial rows of Section 5.2 record that restriction modulo  $K_E$ . For a threefold with a single degree-7 cone point, Theorem 6.1 obstructs smoothing when the root line maps injectively to  $H_2(\hat{X}; \mathbb{Q})$ , as stated in Corollary 9.1. Gross’s primitive contractions have  $\mathbb{Q}$ -factorial targets, for which the root line instead maps to zero. The following proposition explains the distinction.

**Proposition 8.4.** *Let  $X$  be  $\mathbb{Q}$ -factorial and let  $\pi: \hat{X} \rightarrow X$  be a projective crepant resolution. Suppose every irreducible component  $E_i$  of  $\text{Exc}(\pi)$  is contracted to a point and the points  $\pi(E_i)$  are pairwise distinct. Then the rows of Section 5.2 at each of the corresponding divisorial germs vanish identically.*

*Proof.* Because  $X$  is  $\mathbb{Q}$ -factorial and  $\pi$  is projective and birational, the exceptional locus of  $\pi$  has pure codimension one [KM98, Cor. 2.63]. Thus its irreducible components  $E_1, \dots, E_r$  are divisors. By hypothesis they lie over distinct points, as they do for the divisorial germs of this paper.

$\mathbb{Q}$ -factoriality also makes  $\pi^*$  defined on  $\mathrm{Cl}(X)_{\mathbb{Q}}$ , and then  $\mathrm{Cl}(\widehat{X})_{\mathbb{Q}} = \pi^* \mathrm{Cl}(X)_{\mathbb{Q}} \oplus \bigoplus_i \mathbb{Q}E_i$ , with  $\mathrm{Cl}(\widehat{X}) = \mathrm{Pic}(\widehat{X})$  since  $\widehat{X}$  is smooth. Fix  $i$  and write  $E = E_i$ . A pullback  $\pi^*D$  restricts trivially to  $E$  because  $\pi(E)$  is a point, and  $E_j|_E = 0$  for  $j \neq i$  because  $E_j$  and  $E$  lie over distinct points, so the image of  $\mathrm{Pic}(\widehat{X})_{\mathbb{Q}} \rightarrow \mathrm{Pic}(E)_{\mathbb{Q}}$  is spanned by  $E|_E$ . That class is  $K_E$ : adjunction gives  $K_E = (K_{\widehat{X}} + E)|_E$ , and  $K_{\widehat{X}}|_E = (\pi^*K_X)|_E = 0$  by crepancy and the same observation. Every divisor class therefore pairs to zero against  $K_E^\perp$ , and the rows, which are those pairings, vanish.  $\square$

Under the hypotheses of Proposition 8.4, the rows at every divisorial germ are zero, so the kernel contains its entire  $K^\perp$  and the test imposes no obstruction there. In the primitive type II case with exceptional divisor  $\mathrm{dP}_6$  or  $\mathrm{dP}_7$ , [Gro97, Theorem 5.8] supplies the smoothing. The proposition also says that a  $\mathbb{Q}$ -factorial  $X$  admits no *projective* crepant resolution with an exceptional curve; it does not say that  $X$  has no nodes, and indeed a general one-nodal quintic threefold is factorial. Contrapositively:

**Corollary 8.5.** *If the block of rows at a divisorial germ has positive rank, then  $X$  is not  $\mathbb{Q}$ -factorial. Since primitivity in [Gro97, Theorem 5.8] supplies  $\mathbb{Q}$ -factoriality of the target, that theorem, like [Nam97, Corollary 6], is silent on every  $X$  carrying such a germ, and the statements never disagree.*  $\square$

Here  $\mathbb{Q}$ -factoriality is algebraic: every algebraic Weil divisor on  $X$  is  $\mathbb{Q}$ -Cartier, or equivalently  $(\mathrm{Cl}(X)/\mathrm{Pic}(X)) \otimes \mathbb{Q} = 0$ . This must be distinguished from  $\mathbb{Q}$ -factoriality of the analytic germ. The analytic cone germs here are not  $\mathbb{Q}$ -factorial: at a cone over a del Pezzo surface  $E$  the analytic local class group is  $\mathrm{Pic}(E)/\mathbb{Z}K_E$ , of rank  $\rho(E) - 1$ , which is Namikawa's invariant  $\sigma$  [Nam97]. For the toric del Pezzos of the framework that rank is also  $\dim T^1 = n - 3$  (Section 8.2). In general Namikawa proves  $\mu + \sigma \leq \dim T^1 \leq 2\mu + \sigma$  for isolated rational Gorenstein threefold singularities [Nam97, Theorem 1], where  $\mu$  is the dimension of the cokernel of  $d \log: H^1(\widetilde{Y}, \mathcal{O}^*) \otimes \mathbb{C} \rightarrow H^1(\widetilde{Y}, \Omega^1)$  on a good resolution  $\widetilde{Y}$  (it is not the Milnor number), so  $\sigma$  and  $\dim T^1$  can differ only when  $\mu > 0$ . A threefold can be algebraically  $\mathbb{Q}$ -factorial while its analytic germ is not: the target of a primitive type II contraction is the example that matters here, and a general one-nodal quintic is the familiar one.

Thus a positive block rank witnesses the failure of  $\mathbb{Q}$ -factoriality, whereas the relation kernel measures cancellation among the local classes. The two quantities are not interchangeable: on  $X_7$  the block rank is 2 while the relation kernel is zero. On  $X^\circ$  every block has the maximal rank, 2 at each  $\mathrm{dP}_7$  point and 3 at each  $\mathrm{dP}_6$  point.

For admissible polytopes every divisorial block has positive rank. Indeed, the exceptional divisor  $E$  over a singular 2-face is a toric surface whose boundary divisors are the restrictions of the ambient toric divisors, so  $\mathrm{Pic}(\mathbb{P}_{\widehat{\Sigma}})_{\mathbb{Q}} \rightarrow \mathrm{Pic}(E)_{\mathbb{Q}}$  is surjective and the block at that germ has the full rank  $n - 3$  of Proposition 8.2, which is 2 at a pentagon and 3 at a hexagon. Thus no admissible  $X$  carrying a pentagonal or hexagonal face is  $\mathbb{Q}$ -factorial; a node contributes no exceptional divisor to which this argument could be applied. The census bears this out at every one of the 77 polytopes of Proposition 9.2 and at  $\Delta_{19}$ ,  $\Delta_{20}$  and  $X^\circ$ , but it is not the evidence for it. Proposition 8.4 therefore functions here as a consistency check against [Gro97, Theorem 5.8] rather than as a discriminating invariant. In the purely nodal case, a  $\mathbb{Q}$ -factorial threefold has no small projective resolution. On a small analytic resolution of a general one-nodal quintic, the exceptional curve class vanishes in  $H_2$ , so the entire local line belongs to the relation kernel and the necessity test does not obstruct smoothing (Section 9). The mixed criterion uses  $K^\perp$  in place of this local curve class.

## 9. THE CRITERION AS A FINITE TEST

**9.1. The test.** Theorem 6.1 is a finite test for any concrete candidate: compute the matrix of Section 5.2 from the polytope (node rows;  $K^\perp$  bases of the star fans of the pentagon and hexagon faces), restrict to the branch subspaces, and check whether a coordinate required to be nonzero by Theorem 6.1 is forced to vanish. A forced node coordinate contradicts Lemma 3.1; a forced  $\mathrm{dP}_7$  root coordinate contradicts Lemma 3.3; a forced  $s_3$ -coordinate contradicts Lemma 3.3 on profiles through the line component. The test uses nothing about the example beyond Batyrev regularity and its face inventory.

Applied across the admissible polytopes of Definition 8.1 the test has a wide range, and the two ends of that range are worth separating. At the small end it is nearly immediate: with a single germ the criterion asks only whether the germ's root subspace survives in  $H_2(\widehat{X}; \mathbb{Q})$ , which is a two-row rank computation. That case is Section 9.2, and Section 9.3 shows it is not isolated: among admissible polytopes with at most nine vertices carrying a pentagonal 2-face, seventy-six of seventy-seven are obstructed, sixty-seven of them with one singular point. The interest of  $X^\circ$  is that it is at the other end: thirty germs of three kinds, both  $\mathrm{dP}_6$  branch profiles, and a  $36 \times 26$  matrix of rank 21 whose 15-dimensional kernel forces nothing at a node, and where the four coordinates that decide the question appear once the  $\mathrm{dP}_7$  branch restriction is imposed. Sections 9.4 and 9.5 sit between the two, one where the obstruction is carried entirely by the divisorial direction and one where the test is silent.

**9.2. A one-germ illustration.** With one singular point, the criterion reduces to whether the local root line survives in the homology of the crepant resolution.

**Corollary 9.1.** *Let  $X$  be as in Theorem 6.1 with exactly one singular point  $p$ , an anticanonical  $\mathrm{dP}_7$  cone, and suppose the  $(-2)$ -line  $\mathcal{R}_p \subset H_2(L_p; \mathbb{Q})$  maps injectively into  $H_2(\widehat{X}; \mathbb{Q})$ . Then  $X$  admits no one-parameter smoothing.*

*Proof.* With one singular point  $K(X) = \{\kappa \in \mathcal{R}_p : \kappa \mapsto 0 \text{ in } H_2(\widehat{X}; \mathbb{Q})\}$ , which is 0 by hypothesis. A one-parameter smoothing would produce, by Theorem 6.1(2), a  $\kappa \in K(X)$  whose  $p$ -component is a nonzero element of  $\mathcal{R}_p$ .  $\square$

The hypothesis is a finite check: it suffices that the  $(-2)$ -row of Section 5.2 be nonzero, since a class pairing nontrivially with an ambient divisor does not die in  $H_2(\widehat{X}; \mathbb{Q})$ .

Such an  $X$  exists at the seven-vertex bottom of the pentagonal census. Let

$$\begin{aligned} \Delta_7 = \operatorname{conv}\{ & (1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (-6, -4, -1, 0), \\ & (0, 0, 0, 1), (-6, -4, 0, -1), (-3, -2, 1, -1)\}, \end{aligned}$$

reflexive with seven vertices, seven facets and sixteen edges, all of lattice length one. Of its sixteen 2-faces, fifteen are unimodular triangles and one is a pentagon with a single interior lattice point and dual edge of lattice length one, so  $\Delta_7$  is admissible and  $X_7 \subset \mathbb{P}_{\Delta_7}$  has *exactly one* singular point, an anticanonical  $\mathrm{dP}_7$  cone, which is smoothable as a germ. As in Section 5.1,  $\omega_{X_7} \cong \mathcal{O}_{X_7}$  and  $h^1(\mathcal{O}_{X_7}) = 0$ , and  $\widehat{X}_7$  has  $h^{1,1} = 4$  and  $h^{2,1} = 214$ . The two divisorial rows are linearly independent, so the relation-row block has rank 2, the maximum, the ambient kernel is zero, and Corollary 9.1 applies:  $X_7$  admits no one-parameter smoothing.

Since the block has positive rank, Corollary 8.5 says  $X_7$  is not  $\mathbb{Q}$ -factorial, so [Gro97, Theorem 5.8] does not speak to it and there is no conflict with the smoothability of primitive type II contractions with  $\mathrm{dP}_7$  exceptional divisor. For  $X_7$ , the obstruction requires only the rank calculation for its two divisorial rows. At first order the same data show that every first-order deformation of  $X_7$  is locally trivial at its singular point: by [Wue26e] (see

Section 10) the image of the localization map on first-order deformations is the complexified relation kernel, which is zero here; compare [Nam97, Theorem 10] for a contracted del Pezzo surface of degree 6. Vanishing at first order does not by itself exclude a smoothing whose local classifying arc vanishes to order two or more; the period argument behind Corollary 9.1 applies to arcs of every order and excludes it.

**9.3. A census of the admissible polytopes with a pentagonal 2-face, at up to nine vertices.** As a bounded computational application, the test can be applied to every polytope in a finite part of the Kreuzer–Skarke classification.

**Proposition 9.2.** *Among the 3905789 reflexive 4-polytopes with at most nine vertices in the Kreuzer–Skarke classification [KS00], exactly 77 are admissible and carry a pentagonal 2-face, one pentagon apiece: one at seven vertices, thirteen at eight, sixty-three at nine. No pentagonal 2-face occurs below seven vertices. None of the 77 carries a hexagonal 2-face, so each of the corresponding  $X$  has exactly one  $\mathrm{dP}_7$  cone point and no  $\mathrm{dP}_6$  point; the search records that each has at most two nodes, the inventories being 67 with the cone point alone, 9 with one node in addition, and 1 with two nodes. The test of Theorem 6.1 obstructs all but one of them, so those 76 admit no one-parameter smoothing, and by Corollary 6.7 they are not smoothable over any base germ. On the remaining polytope, which has nine vertices and two nodes, the test is silent.*

*Proof.* Each such  $X$  satisfies the standing hypotheses of Theorem 6.1 by Section 8.1, with the crepant resolution chosen there: the restriction of a maximal projective crepant partial desingularization of  $\mathbb{P}_{\Pi}$  that refines the star subdivision at the interior point of the pentagon, small over the nodes, and the blowup of the vertex, with exceptional surface  $\mathrm{dP}_7$ , at the cone point. The kernel calculations of Section 9.1 are recomputed for each of the 77 from its vertex list (Appendix A). Since no  $\mathrm{dP}_6$  point occurs, the only branch restriction is  $\alpha = 0$  at the pentagon, and a forced node or  $(-2)$ -coordinate is contradicted by Lemma 3.1 or Lemma 3.3.  $\square$

The enumeration itself is the admissibility-restricted extract of a scan reported in the companion paper [Wue26e]. The input files are the at-most-nine-vertex part of the complete database copy verified in [Wue26a, §6.1]: every entry is reflexive and the normal forms are pairwise distinct up to lattice equivalence. The displayed input count is read from these files. The germ inventory and the matrix are recomputed here from each stored vertex list. The 67 single-germ cases are instances of Corollary 9.1. For these cases and the 9 cases with one node in addition, the full matrix has zero relation kernel, so the required relation cannot exist. Together with the distinguished example of Section 6, the census therefore gives seventy-six further nonsmoothable threefolds, none of them  $X^\circ$ ,  $X_{20}$  or  $X_{19}$ , whose polytopes have 26, 20 and 19 vertices and lie outside it.

The enumeration also identifies the single polytope among the 77 for which the obstruction vanishes. The companion paper [Wue26e] proves that its hypersurface is smoothable.

The census is restricted to polytopes with a pentagonal 2-face, and that restriction is not vacuous: admissible polytopes in the same range carry hexagonal 2-faces, hence  $\mathrm{dP}_6$  cone points. For example, an eight-vertex case is obtained from  $\Delta_7$  by adjoining  $(-3, -2, -1, 1)$ . On such an  $X$  Theorem 6.1 constrains the one-parameter  $\mathrm{dP}_6$  branch and says nothing about the  $A_2$  branch, so the conclusions above do not extend to them and we do not claim them.

**9.4. An obstruction the nodal theory cannot see.** The test also obstructs threefolds that no purely nodal argument can reach. Let  $\Delta_{20}$  be the 20-vertex reflexive polytope

$$\begin{aligned} \text{conv}\{ & (1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1), (1, -1, -1, -1), (-1, 1, 1, 1), (0, 0, 0, -1), \\ & (0, 0, -1, 0), (0, -1, 0, 0), (-1, 1, 0, 1), (0, 0, -1, -1), (-1, 1, 0, 0), (-1, 0, 1, 1), (0, -1, 0, -1), \\ & (-1, 0, 1, 0), (0, -1, -1, -1), (-1, 0, 0, 1), (-1, 0, -1, 0), (-1, -1, 0, 0), (-1, -1, 1, 0)\}, \end{aligned}$$

one of the both-sides-unit polytopes classified in [Wue26c], where it appears in the accompanying data. Its generic anticanonical hypersurface  $X_{20}$  has seventeen ordinary double points and one  $\text{dP}_7$ -cone point, all isolated; as in Section 5.1 it satisfies  $\omega_{X_{20}} \cong \mathcal{O}_{X_{20}}$  and  $h^1(\mathcal{O}_{X_{20}}) = 0$ , and a smooth projective crepant subdivision exists, obtained as in Section 5.2 from the seventeen square diagonals and the star subdivision at the interior point of the pentagon. Here  $\Delta_{20}$  has 22 lattice points and no facet with an interior point, so Batyrev's formulas give  $h^{1,1}(\widehat{X}_{20}) = 22 - 5 = 17 = \text{rank Pic}$  of the ambient toric fourfold: the ambient divisor classes span  $H^2(\widehat{X}_{20}; \mathbb{Q})$ , and the ambient kernel of Section 5.3 coincides with the topological one rather than merely containing it, as Proposition 8.3 also gives. The outcome is:

- (1) the nodal subsystem has rank 11 of 17 and *no coloop*, so, the two kernels agreeing, every exceptional curve of a small resolution does appear with nonzero coefficient in some relation among them: Friedman's necessary condition is satisfied;
- (2) the mixed matrix has 19 rows on 21 rays, its kernel has dimension 6, and *both*  $\text{dP}_7$  coordinates vanish identically on it, the root coordinate among them.

By Theorem 6.1(2) and Lemma 3.3,  $X_{20}$  admits no one-parameter smoothing. Here the obstruction is carried entirely by the divisorial direction, and a reader applying [Fri86, RT09] alone would find nothing wrong. On  $X^\circ$ , by contrast, the two nodal coloops already signal that the nodal theory is insufficient.

**9.5. Where the test is silent.** We also ran the test on the anticanonical hypersurface  $X_{19}$  of the 19-vertex reflexive polytope  $\Delta_{19}$ , whose generic member has fourteen nodes and one  $\text{dP}_7$ -cone point, all isolated. The present necessity test is silent on this example, which the companion paper [Wue26e] proves smoothable. The vertices of  $\Delta_{19} \subset N \cong \mathbb{Z}^4$  are

$$\begin{aligned} & (1, 0, 0, 0), \quad (0, 1, 0, 0), \quad (0, 0, 1, 0), \quad (0, 0, -1, 0), \\ & (0, -1, 0, 0), \quad (1, -1, -1, 0), \quad (0, 0, 0, 1), \quad (-1, 1, 1, -1), \\ & (0, -1, 0, 1), \quad (0, 0, -1, 1), \quad (-1, 1, 0, -1), \quad (-1, 0, 1, -1), \\ & (-1, 0, 1, 0), \quad (-1, 1, 0, 0), \quad (0, -1, -1, 0), \quad (0, -1, -1, 1), \\ & (-1, 0, 0, -1), \quad (-2, 1, 1, -1), \quad (-1, 0, 0, 1). \end{aligned}$$

It is the polytope of [Wue26c, Section 5.3]. The outcome is structurally opposite to  $X^\circ$ :

- (1) the node subsystem has rank 12 with two coloops: as for  $X^\circ$ , no purely nodal mechanism can smooth  $X_{19}$ ;
- (2) on the full kernel (dimension 3) the complementary  $\text{dP}_7$ -coordinate  $\alpha$  is already forced to zero, so every relation class lies on the  $(-2)$ -line without imposing the branch restriction; and
- (3) no coordinate required by Theorem 6.1 is forced to vanish: a generic kernel class is nonzero in all fourteen node coordinates and in the  $(-2)$ -coordinate simultaneously.

The necessity theory of this paper is therefore silent on  $X_{19}$ , and the companion paper [Wue26e] proves the required mixed smoothing exists. As a control in the opposite direction, the test applied to the generic one-nodal quintic threefold is silent as well, as

it must be: there the exceptional curve class vanishes in  $H_2$  of the small resolution (the threefold is  $\mathbb{Q}$ -factorial), the kernel is the whole line  $H_2(L; \mathbb{Q})$ , and the node coordinate is realized, the smoothable side of the nodal picture. Every smoothing of  $X_{19}$  must use the  $\mathrm{dP}_7$  direction. The companion paper [Wue26e] proves its existence by the mixed deformation criterion and also studies the limitations of deformations induced by a toric ambient space.

## 10. DISCUSSION

**10.1. The two-parameter degree-six smoothing component.** Theorem 6.1 treats  $\mathrm{dP}_6$  points smoothed through the one-parameter component. The companion paper [Wue26e] treats the two-parameter component by an explicit simultaneous partial resolution that replaces the central cone by three nodes and is an isomorphism on every noncentral fiber. In the marking

$$R_{A_2} = \{\mu_1 e_1 + \mu_2 e_2 + \mu_3 e_3 : \mu_1 + \mu_2 + \mu_3 = 0\},$$

the required condition is  $\mu_1 \mu_2 \mu_3 \neq 0$ . It is stronger than requiring a nonzero vector in  $R_{A_2}$  and holds for arbitrary smoothing arcs. None of the proofs in the present paper require that extension.

**10.2. First-order theory.** The companion paper [Wue26e] identifies the image of the localization map on first-order deformations with the topological relation kernel, using the local-to-global comparison of Friedman–Laza [FL25]. Under Proposition 8.3, this is the matrix kernel used here. The period argument of the present paper gives necessity directly for an actual smoothing arc and does not assume that its first derivative smooths each germ. In the purely nodal case, the criterion for actual smoothings is already [RT09, Thm. 1.3].

**10.3. Sufficiency.** The companion paper [Wue26e, Thms. A–B] proves an exact converse for connected normal projective threefolds with trivial dualizing sheaf,  $H^1(X, \mathcal{O}_X) = 0$ , and the prescribed analytic germs: nodes and anticanonical cones over smooth del Pezzo surfaces of degrees five, six and seven, or over  $\mathbb{P}^1 \times \mathbb{P}^1$ . For each choice of local components, the relation kernel must avoid every local discriminant: a nonzero coefficient in each one-dimensional local summand, three nonzero coefficients in each  $A_2$  summand, and five nonzero conic-pencil pairings at each degree-five point. Thus silence of the test in the present paper gives the converse only when it includes all the nonvanishing conditions for the chosen components.

The proof replaces the degree-six and degree-seven cones by proper crepant partial models whose full local deformation bases are smooth. Friedman’s generalized  $T^1$  lifting theorem, together with marked local lifting, makes the partial model unobstructed; analytic blow-down then gives smoothness of the selected base of  $X$  whenever its relation kernel projects nontrivially to every degree-six summand. No  $\mathbb{Q}$ -factoriality assumption is imposed on  $X$ , and the partial model need not be projective. The criterion also proves the smoothability stated in Example 1.3. The separate issue of which deformations can be induced by a toric ambient family is studied in [Wue26e].

**10.4. Mirror symmetry.** For the admissible hypersurfaces of Definition 8.1, the companion paper [Wue26f] studies the relation lattice on the mirror side. For each profile in the sense of Section 5.2 it passes to a nodal crepant partial resolution induced by a projective toric morphism. It identifies the integral relations among the exceptional curves of a small resolution with the relations among the vanishing three-spheres of suitable degenerations of the Batyrev mirror family, and it characterizes smoothability by relations among those spheres.

## APPENDIX A. COMPUTATIONAL CERTIFICATES

The exact lattice and matrix claims used in the proofs are reproduced by the Python programs listed below, in exact rational arithmetic and independently of any computer-algebra system. A third self-contained program combines exact integral checks with a floating-point diagnostic of the period parametrization, whose underlying identity is proved exactly in Lemma 3.2. The inputs are Tables 1 and 2, the ancillary subdivision and restriction data documents, the polytope of [Wue26c, Section 5.3], and the vertex lists of the 77 polytopes of Proposition 9.2:

- the lattice package (`milnor_kernels.py`): the boundary structure of the exceptional surfaces, the norms and pairings of the  $K^\perp$  rows, the root decompositions, the  $(-2)$ -enumerations with rigorous search bounds, and the isometries to the standard del Pezzo markings;
- the matrix package (`global_kernel.py`): the reconstruction of all 36 rows from the printed data, the ranks, coloops, kernels, the four profiles and the enlarged profile with their forced-zero lists, and the criticality statement of Section 5.3(3);
- the period package (`dp_periods.py`): the exact fan combinatorics behind (2) (the  $\pm 1$  pairings on all rays, the divisor computation, the slice self-intersection patterns), the corner-chart transitions, and the constancy of the pullback in Lemma 3.2, verified to machine precision at random points;
- `examples.py`, the general implementation of Section 9, which recomputes every number of Section 5.3 for  $X^\circ$  from the polytope, and the  $\Delta_{19}$  and  $\Delta_{20}$  computations, including the absence of nodal coloops on  $\Delta_{20}$ ;
- the one-germ package (`lone_germ.py`): the reflexive 4-polytope  $\Delta_7$  of Section 9.2, its face inventory and admissibility, the rank-2 relation-row block and zero kernel, and the Hodge numbers  $(4, 214)$ ;
- the census package (`classification.py`, reading the extract `framework_77.json`): the germ inventory, the matrix and the verdict for each of the 77 polytopes of Proposition 9.2, rebuilt from the stored vertex lists;
- the relation-row package (`relation_class.py`): for  $X^\circ$ ,  $X_{20}$ ,  $X_{19}$  and  $X_7$ , that every relation row satisfies  $\sum_j a_j v_j = 0$  and  $\sum_j a_j = 0$ , carries coefficient zero at the interior lattice point, and that the rows at a germ span the whole ray-relation space of rank  $n - 3$  (Proposition 8.2); together with the per-germ block ranks of Corollary 8.5 and an independent reconstruction of rank  $B = 21$  for  $X^\circ$ ;
- the kernel-equality package (`kernel_equality.py`): the vanishing of both Batyrev correction terms, and the reason each 2-face contributes zero, for  $\Delta^\vee$ ,  $\Delta$ ,  $\Delta_{20}$ ,  $\Delta_{19}$ ,  $\Delta_7$  and all 77 census polytopes (Proposition 8.3);
- the mirror-partner package (`mirror_partner.py`): the reconstruction of the twenty-four rows of  $B_\Delta$  from Tables 3 and 4, including the derivation of both exceptional surface lattices from the face data of Table 3, the reduced-rigidity of the  $\mathbb{F}_1$  germ by the zero-sum test on its edge vectors, the two vanishing columns of item (3), the root decomposition  $A_1 \oplus A_2$  of  $K^\perp(\text{dP}_6)$ , the absence of  $(-2)$ -classes in  $K^\perp(\mathbb{F}_1)$  and the square  $-8$  of its generator, and the ranks, coloops and forced-zero lists of Section 7.

The fixed subdivision of Section 5.2 and its smoothness and projectivity are certified by an exact integral support function; the Batyrev regularity of the reference equation of  $X^\circ$ , including nondegeneracy along every face, is certified at an integral witness (`resolution_fan.sage` and `global_section.sage`, with five further Sage certificates and the face-data module

`fixed_example.py`). The subdivision of Section 7, its 94 unimodular maximal cones, the two stars and the twenty diagonals are certified the same way by `mirror_partner_fan.sage`.

The arXiv ancillary bundle includes this directory together with the shared exact toric modules used by the programs. The same files are public, together with the programs and data of the companion papers, at

<https://github.com/bwuebben/calabi-yau-smoothability>.

## REFERENCES

- [Alt95] K. Altmann, *Minkowski sums and homogeneous deformations of toric varieties*, Tôhoku Math. J. (2) **47** (1995), 151–184. (Its §6 grew out of the preprint *Toric  $\mathbb{Q}$ -Gorenstein singularities*, arXiv:alg-geom/9403003, where the statement cited here as (6.5) is (2.9), Case 2; it is restated as [Alt97, (6.3)].)
- [Alt97] K. Altmann, *The versal deformation of an isolated toric Gorenstein singularity*, Invent. Math. **128** (1997), 443–479. (For the proof of the component description see also arXiv:alg-geom/9403004v2, (8.2)–(8.3).)
- [Art68] M. Artin, *On the solutions of analytic equations*, Invent. Math. **5** (1968), 277–291.
- [Art69] M. Artin, *Algebraic approximation of structures over complete local rings*, Publ. Math. IHÉS **36** (1969), 23–58.
- [Bat94] V. V. Batyrev, *Dual polyhedra and mirror symmetry for Calabi–Yau hypersurfaces in toric varieties*, J. Algebraic Geom. **3** (1994), 493–535.
- [BM88] E. Bierstone and P. D. Milman, *Semianalytic and subanalytic sets*, Publ. Math. IHÉS **67** (1988), 5–42.
- [Con00] B. Conrad, *Grothendieck duality and base change*, Lecture Notes in Math. **1750**, Springer, 2000.
- [dJvS94] T. de Jong and D. van Straten, *On the deformation theory of rational surface singularities with reduced fundamental cycle*, J. Algebraic Geom. **3** (1994), 117–172.
- [DK86] V. I. Danilov and A. G. Khovanskii, *Newton polyhedra and an algorithm for computing Hodge–Deligne numbers*, Izv. Akad. Nauk SSSR Ser. Mat. **50** (1986), 925–945; English transl. in Math. USSR-Izv. **29** (1987), 279–298.
- [DLRS10] J. A. De Loera, J. Rambau and F. Santos, *Triangulations: Structures for Algorithms and Applications*, Algorithms and Computation in Mathematics **25**, Springer, Berlin, 2010.
- [Fri86] R. Friedman, *Simultaneous resolution of threefold double points*, Math. Ann. **274** (1986), 671–689.
- [Fri26] R. Friedman, *Unobstructed deformations for singular Calabi–Yau varieties*, arXiv:2506.09857v3, 13 February 2026.
- [FL24] R. Friedman and R. Laza, *Deformations of some local Calabi–Yau manifolds*, Épijournal Géom. Algébrique, volume in honour of C. Voisin (2024), Art. No. 18; arXiv:2203.11738.
- [FL25] R. Friedman and R. Laza, *Deformations of singular Fano and Calabi–Yau varieties*, J. Differential Geom. **131** (2025), 65–131.
- [Fuj80] T. Fujita, *On the structure of polarized manifolds with total deficiency one, I*, J. Math. Soc. Japan **32** (1980), 709–725.
- [GKZ94] I. M. Gelfand, M. M. Kapranov and A. V. Zelevinsky, *Discriminants, Resultants, and Multidimensional Determinants*, Birkhäuser, Boston, 1994.
- [Gra60] H. Grauert, *Ein Theorem der analytischen Garbentheorie und die Modulräume komplexer Strukturen*, Publ. Math. IHÉS **5** (1960), 5–64.
- [Gra72] H. Grauert, *Über die Deformation isolierter Singularitäten analytischer Mengen*, Invent. Math. **15** (1972), 171–198.
- [Gra74] H. Grauert, *Der Satz von Kuranishi für kompakte komplexe Räume*, Invent. Math. **25** (1974), 107–142.
- [Gre20] G.-M. Greuel, *Deformation and smoothing of singularities*, in: Handbook of Geometry and Topology of Singularities I, Springer, 2020; arXiv:1903.03661.
- [Gro97] M. Gross, *Deforming Calabi–Yau threefolds*, Math. Ann. **308** (1997), 187–220.
- [IV12] N. Ilten and R. Vollmert, *Deformations of rational  $T$ -varieties*, J. Algebraic Geom. **21** (2012), 473–493.
- [KP21] A.-S. Kaloghiros and A. Petracci, *On toric geometry and  $K$ -stability of Fano varieties*, Trans. Amer. Math. Soc. Ser. B **8** (2021), 548–577.

- [KS72] A. Kas and M. Schlessinger, *On the versal deformation of a complex space with an isolated singularity*, Math. Ann. **196** (1972), 23–29.
- [KS00] M. Kreuzer and H. Skarke, *Complete classification of reflexive polyhedra in four dimensions*, Adv. Theor. Math. Phys. **4** (2000), 1209–1230.
- [KP23] A. Kuznetsov and Yu. Prokhorov, *On higher-dimensional del Pezzo varieties*, Izv. Math. **87** (2023), no. 3, 488–561.
- [Loj65] S. Łojasiewicz, *Ensembles semi-analytiques*, lecture notes, IHÉS, Bures-sur-Yvette, 1965.
- [Mav04] A. R. Mavlyutov, *Deformations of Calabi–Yau hypersurfaces arising from deformations of toric varieties*, Invent. Math. **157** (2004), 621–633.
- [Mil68] J. Milnor, *Singular points of complex hypersurfaces*, Ann. of Math. Studies **61**, Princeton, 1968.
- [Pet19] A. Petracci, *On deformations of toric Fano varieties*, arXiv:1912.01538.
- [Pet20] A. Petracci, *An example of mirror symmetry for Fano threefolds*, in: Birational Geometry and Moduli Spaces, Springer INdAM Ser. **39** (2020), 173–188.
- [Pin74] H. Pinkham, *Deformations of algebraic varieties with  $G_m$  action*, Astérisque **20** (1974).
- [Nam02] Y. Namikawa, *Stratified local moduli of Calabi–Yau threefolds*, Topology **41** (2002), no. 6, 1219–1237.
- [Nin26] H. Ning, *Deformations, local freeness, and base change for higher Du Bois singularities*, preprint, 2026; arXiv:2608.07849.
- [Nam97] Y. Namikawa, *Deformation theory of Calabi–Yau threefolds and certain invariants of singularities*, J. Algebraic Geom. **6** (1997), 753–776. (Not to be confused with the author’s *Smoothing Fano 3-folds*, ibid. **6** (1997), 307–324, which some sources also cite as [Nam97].)
- [Agu26] R. Aguilar, *Log conifold transitions*, preprint, 2026; arXiv:2606.31072.
- [Col25] T. C. Collins, *An introduction to conifold transitions*, lecture notes, 2025; arXiv:2509.01002.
- [KM98] J. Kollár and S. Mori, *Birational geometry of algebraic varieties*, Cambridge Tracts in Mathematics 134, Cambridge University Press, Cambridge, 1998.
- [PP15] P. Popescu-Pampu, *On the smoothings of non-normal isolated surface singularities*, J. Singul. **12** (2015), 164–179.
- [RT09] S. Rollenske and R. Thomas, *Smoothing nodal Calabi–Yau  $n$ -folds*, J. Topol. **2** (2009), 405–421.
- [Wue26a] B. J. Wuebben, *Non-smoothable Calabi–Yau threefolds from reflexive polytopes*, preprint, 2026; arXiv:2609.28499.
- [Wue26b] B. J. Wuebben, *Deformations of toric pairs and the smoothing of Batyrev Calabi–Yau threefolds*, preprint, 2026; arXiv:2610.06884.
- [Wue26c] B. J. Wuebben, *Doubly isolated Batyrev mirror pairs and non-smoothable Calabi–Yau threefolds*, preprint, 2026; arXiv:2610.03753.
- [Wue26e] B. J. Wuebben, *Smoothing Calabi–Yau threefolds with nodes and del Pezzo cone points*, preprint, 2026; available at <https://github.com/bwuebben/calabi-yau-smoothability>.
- [Wue26f] B. J. Wuebben, *Mirror symmetry for conifold relations in Calabi–Yau threefolds*, preprint, 2026; available at <https://github.com/bwuebben/calabi-yau-smoothability>.

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